



API Specification

Public Transportation Open Data Center : API Specification

Association for Open Data of Public Transportation – info@odpt.org – Version 4.14, 2024-03-27

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1. Overview

The Public Transportation Open Data Center API (hereafter API) provides information on railways, buses and air

flights.

1.1. Overview

Public transportation handled by this API refers to transportation used by an unspecified number of people. Specifically, this API provides information on railways, buses and air flights.

Data used in this API is provided in ucR format based on the concept of semantics, specifically in JSON-LD format. First, an identifier (ucode) is assigned to an existing concept (entity). Taking a railway as an example, the entities are

- train
- station
- line
- station timetable
- operation information

and so on. Then, the relationships (Relation, Link) existing between these entities are described.

A relationship between real entities expressed in this manner by three elements of a subject, a predicate, and an object is called a triple.

All information provided by this API is described based on such triples.

Then, we have generalized the required properties for each entity that exists as public transportation information, and defined them as RDF classNameees. The details are described in the next section.

NOTE

- Reference Links
- uID Center
 - W3C Resource Description Framework (RDF)
 - JSON-LD.org

1.1.1. Data type names and their definitions

Type	rdf:type	Name	Details
Common	odpt:Calendar	Day of the week / date classNameification	Describes day of the week / date classNameification
Common	odpt:Operator	Operator	Describes public transportation operators
Railway	odpt:Station	Station information	Describes information related to railway stations, such as the name and location of the station
Railway	odpt:StationTimetable	Station timetable	Timetable information of trains leaving and arriving at the station
Railway	odpt:Train	Train information	Describes real-time train information, such as train location and destination
Railway	odpt:TrainTimetable	Train timetable	Describes when a train arrives at which station and when it departs from it
Railway	odpt:TrainInformation	Operation information	Describes the real-time operation status of railway lines
Railway	odpt:TrainType	Train type	Defines the type of train, such as local or rapid
Railway	odpt:RailDirection	Direction of travel	Defines the direction of travel of the train, such as going up or down
Railway	odpt:Railway	Line information	Describes lines and service routes of trains
Railway	odpt:RailwayFare	Fare information	Describes railway fare
Railway	odpt:PassengerSurvey	Number of people getting on and off at each station	Describes the result of counting the number of people getting on and off at each station

Bus	odpt:Bus	Bus information	Describes real-time information of the bus, such as the location and destination
Bus	odpt:BusTimetable	Bus timetable	Describes when a bus arrives at a bus stop, and departs from it. E.g. route timetable, staff, and operation table
Bus	odpt:BusroutePattern	Bus service line information	Describes bus line information
Bus	odpt:BusroutePatternFare	Bus fare information	Describes bus fare
Bus	odpt:BusstopPole	Bus stop post information	Describes signpost information at bus stops
Bus	odpt:BusstopPoleTimetable	Bus stop post timetable	Describes when a bus arrives at a bus stop and departs from it. E.g. bus stop timetable, post timetable, and post
Air flights	odpt:Airport	Airport information	Describes information related to an airport, such as the name and location of the airport
Air flights	odpt:AirportTerminal	Airport terminal information	Describes information related to the airport terminal, such as the name of the airport terminal
Air flights	odpt:FlightInformationArrival	Flight arrival information	Describes real-time information of the flights arriving at the airport on the day
Air flights	odpt:FlightInformationDeparture	Flight departure information	Describes real-time information of the flights departing from the airport on the day
Air flights	odpt:FlightSchedule	Flight timetable	Describes scheduled time of flight arrival/departure. E.g. monthly timetable and weekly timetable
Air flights	odpt:FlightStatus	Flight status	Defines the status of flights arriving and departing at the airport, such as boarding and departed

Table 1. Names of data types to be provided and their definitions

1.2. Data to be provided

The following describes the names of data types provided by this API and their definitions.

1.2.1. Namespace

The following are used as the namespaces of the RDF vocabulary set handled by this API.

Namespaces	Identifier	Remarks
rdf	http://www.w3.org/1999/02/22-rdf-syntax-ns#	RDF
dc	http://purl.org/dc/elements/1.1/	Dublin Core
geo	http://www.opengis.net/ont/geosparql#	OGC GeoSPARQL Geographic information description vocabulary
owl	http://www.w3.org/2002/07/owl#	W3C Web Ontology Language
xsd	http://www.w3.org/2001/XMLSchema-datatypes	XML Schema Datatypes
ug	http://uidcenter.org/vocab/ucr/ug#	Vocabulary set for uID Center features
ugsrv	http://uidcenter.org/vocab/ucr/ugsrv#	Vocabulary set for uID Center Geographic Information Service

Table 2. Namespaces

1.3. Interface

This API provides an interface that uses HTTPS RESTful API.

Data acquisition / search API

API that returns information that matches the property specified by the query parameter

Feature information acquisition / search API

API that provides a filtering function by geographic area specified by geographic information

1.3.1. Note

The points to note in this API document are described below.

- All data handled by this API return values expressed in JSON-LD.
- Each data is expressed in Object format.
- All results output by the API are returned as an Array of Objects.
 - If no match is found, a zero-length Array (empty Array) is returned.
 - If there is only one matching result, an array of length 1 (Array with one Hash) is returned.
- Query parameters of a request must be in URI encoding format.
 - For example, "東京" needs to be encoded as "%E6%9D%B1%E4%BA%AC".
 - In order to maintain readability, this document does not show the result of URI encoding.
- At the time of request, API access key (PRIMARY_OR_SECONDARY_KEY) must be always given by the following method.
 - Query parameter `acl:consumerKey=PRIMARY_OR_SECONDARY_KEY`
- If the result output by API exceeds the [upper limit of the system](#), the result truncated below the upper limit is returned.
 - It is necessary to acquire necessary information again after narrowing down by proper query parameter.
- When an ID is returned as `odpt:xxx` in the output result, the name (`dc:title` or `odpt:xxxTitle` of the ID) may be added as `odpt:xxxTitle` and returned also.

1.4. Data retrieval API (/v4/RDF_TYPE?)

`/api/v4/RDF_TYPE` returns information that matches the specified query.

end point

`https://api.odpt.org/api/v4/RDF_TYPE`

HTTP Method

GET

HTTP Status Code

- 200: Successful completion
- 400: Invalid parameter
- 401: `acl:consumerKey` is incorrect
- 403: lacking authority
- 404: No applicable data
- 405: unallowed HTTP Method
- 500: Server internal error
- 503: Service unavailable

Query parameters

Query	Description	Required
<code>rdf:type</code>	Specify the type of data to be acquired. See <code>rdf:type</code> list.	○
<code>acl:consumerKey</code>	Specify an access token issued on the developer site.	○

PREDICATE	Perform filtering by specifying the property of a className specified by rdf:type.	
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PREDICATE is a property name that changes for each rdf:type. The list of properties for each rdf:type is described in the vocabulary specification.

Request example

```
curl -X GET https://api.odpt.org/api/v4/odpt:Train?acl:consumerKey=ACL_CONSUMERKEY
```

Response example

```
[
  {
    "@context": "http://vocab.odpt.org/context_odpt_Train.jsonld",
    "@type": "odpt:Train",
    "@id": "urn:ucode:_00001C00000000000000010000030FD5D6",
    "dc:date": "2017-11-28T11:02:15+09:00",
    "dct:valid": "2017-11-28T11:02:45+09:00",
    "odpt:frequency": 30,
    "odpt:railway": "odpt.Railway:TokyoMetro.Yurakucho",
    "owl:sameAs": "odpt.Train:TokyoMetro.Yurakucho.B1045S",
    "odpt:trainNumber": "B1045S",
    "odpt:trainType": "odpt.TrainType:TokyoMetro.Local",
    "odpt:delay": 0,
    "odpt:originStation": [ "odpt.Station:TokyoMetro.Yurakucho.ShinKiba" ],
    "odpt:destinationStation": [ "odpt.Station:TokyoMetro.Yurakucho.Wakoshi" ],
    "odpt:fromStation": "odpt.Station:TokyoMetro.Yurakucho.ChikatetsuNarimasu",
    "odpt:toStation": "odpt.Station:TokyoMetro.Yurakucho.Wakoshi",
    "odpt:railDirection": "odpt.RailDirection:TokyoMetro.Wakoshi",
    "odpt:operator": "odpt.Operator:TokyoMetro"
  }
]
```

1.4.1. Filtering

By specifying PREDICATE as a parameter, you can filter search results by a specific value.

- If the property has a string or numeric value, specify the property name itself in PREDICATE.
- If a property is an array type where each element of the array type is a character string type or numeric type, specify the property name itself in PREDICATE. In this case, "property=value" means that the value specified on the right side is included in at least one or more elements of the array on the left side.
- If a property is an object type where the value of the object type key is a character string type or numeric type, specify "property.key" in PREDICATE. Similarly, nesting such as "property.keyA.keyB" can be specified.
- Other properties (such as when an array element is an object type) cannot be specified.

For example, to acquire only the information of 東京 Station from the station information (rdf:type is odpt:Station), specify dc:title of odpt:Station in PREDICATE and specify "東京" as its value.

Example request: dc:title=東京

```
curl 'https://api.odpt.org/api/v4/odpt:Station?dc:title=東京&acl:consumerKey=ACL_CONSUMERKEY'
```

Alternatively, odpt:stationTitle.en (English station name) of odpt:Station may be specified in PREDICATE, and "Tokyo" may be specified as the value.

Example request: odpt:stationTitle.en=Tokyo

```
curl 'https://api.odpt.org/api/v4/odpt:Station?odpt:stationTitle.en=Tokyo&acl:consumerKey=ACL_CONSUMERKEY'
```

When specifying multiple search conditions, "OR" search is requested by specifying the value of PREDICATE separated by a comma ",". For example, if you would like to acquire information about 東京 and 五反田 at the same time for station information, specify dc:title with 東京 and 五反田 separated by "," and search.

Example request: dc:title=東京,五反田

```
curl 'https://api.odpt.org/api/v4/odpt:Station?dc:title=東京,五反田&acl:consumerKey=ACL_CONSUMER_KEY'
```

1.5. Data Dump API (/v4/RDF_TYPE.json?)

/api/v4/RDF_TYPE.json returns a dump of the specified data type (RDF_TYPE). The data dump API returns all data stored in each of the following data types.

Type	rdf:type	Name	Details
Common	odpt:Calendar	Day of the week / date classNameification	Describes day of the week / date classNameification
Common	odpt:Operator	Operator	Describes public transportation operators
Railway	odpt:Station	Station information	Describes information related to railway stations, such as the name and location of the station
Railway	odpt:StationTimetable	Station timetable	Timetable information of trains leaving and arriving at the station
Railway	odpt:TrainTimetable	Train timetable	Describes when a train arrives at which station and when it departs from it
Railway	odpt:TrainType	Train type	Defines the type of train, such as local or rapid
Railway	odpt:RailDirection	Direction of travel	Defines the direction of travel of the train, such as going up or down
Railway	odpt:Railway	Line information	Describes lines and service routes of trains
Railway	odpt:RailwayFare	Fare information	Describes railway fare
Railway	odpt:PassengerSurvey	Number of people getting on and off at each station	Describes the result of counting the number of people getting on and off at each station
Bus	odpt:BusTimetable	Bus timetable	Describes when a bus arrives at a bus stop, and departs from it. E.g. route timetable, staffe, and operation table
Bus	odpt:BusroutePattern	Bus service line information	Describes bus line information
Bus	odpt:BusroutePatternFare	Bus fare information	Describes bus fare
Bus	odpt:BusstopPole	Bus stop post information	Describes signpost information at bus stops
Bus	odpt:BusstopPoleTimetable	Bus stop post timetable	Describes when a bus arrives at a bus stop and departs from it. E.g. bus stop timetable, post timetable, and post
Air flights	odpt:Airport	Airport information	Describes information related to an airport, such as the name and location of the airport
Air flights	odpt:AirportTerminal	Airport terminal information	Describes information related to the airport terminal, such as the name of the airport terminal
Air flights	odpt:FlightSchedule	Flight timetable	Describes scheduled time of flight arrival/departure. E.g. monthly timetable and weekly timetable
Air flights	odpt:FlightStatus	Flight status	Defining the status of flights departing from and landing at an airport

Table 3. Data types supported by the data dump API

end point

https://api.odpt.org/api/v4/RDF_TYPE.json

HTTP Method

GET

HTTP Status Code

- 200: Successful completion
- 400: Invalid parameter
- 401: acl:consumerKey is incorrect
- 403: lacking authority
- 404: No applicable data
- 405: unallowed HTTP Method
- 500: Server internal error
- 503: Service unavailable

Query parameters

Query	Description	Required
RDF_TYPE	Specify the type of data to be acquired. See rdf:type list.	☐
acl:consumerKey	Specify an access token issued on the developer site.	☐

Since the data dump API returns an HTTP 301 redirect to the URL to the dump file after the request, be careful when processing.

Request example

```
curl -X GET -L https://api.odpt.org/api/v4/odpt:Station.json?acl:consumerKey=ACL_CONSUMERKEY
```

Response example

```
[
  {
    "@context": "http://vocab.odpt.org/context_odpt.jsonld",
    "@id": "urn:ucode:_00001C000000000000010000031028E6",
    "@type": "odpt:Station",
    "dc:title": "東京",
    "owl:sameAs": "odpt.Station:JR-East.Yamanote.Tokyo",
  },
  {
    "@context": "http://vocab.odpt.org/context_odpt.jsonld",
    "@id": "urn:ucode:_00001C000000000000010000031028E7",
    "@type": "odpt:Station",
    "dc:title": "有楽町",
    "owl:sameAs": "odpt.Station:JR-East.Yamanote.Yurakucho",
  }
]
```

1.6. Data acquisition API (GET /v4/datapoints/\$DATA_URI)

/v4/datapoints/\$DATA_URI fetches data with the URI specified by DATA_URI as ID. The data acquisition API supports the same data types of the data dump API. As the ID that can be specified in \$DATA_URI, ucode starting with urn:ucode:_ or the value of owl:sameAs that has one-tone mapping with ucode can be specified.

end point

[https://api.odpt.org/api/v4/datapoints/\\$DATA_URI](https://api.odpt.org/api/v4/datapoints/$DATA_URI)

HTTP Method

GET

- 200: Successful completion
- 400: Invalid parameter
- 401: acl:consumerKey is incorrect
- 402: Billing required
- 403: lacking authority
- 404: No applicable data
- 405: unallowed HTTP Method
- 500: Server internal error
- 503: Service unavailable

Query	Description	Required
\$DATA_URI	Specify the URI of the data to be acquired	☐
acl:consumerKey	Specify an access token issued on the developer site.	☐

```
curl -X GET 'https://api.odpt.org/api/v4/datapoints/urn:ucode:_00001C00000000000001000003051'
```

```
curl -X GET 'https://api.odpt.org/api/v4/datapoints/odpt.Station:TokyoMetro.Ginza.Ueno?acl:0'
```

In /v4/places, it is possible to narrow down the search area using geographic information. The data that can be searched by the feature information search API are listed below.

Table 4. Data types targeted by the feature information search API

Query	Description	Required
RDF_TYPE	Specify the type of data to be acquired. See rdf:type list.	☐
lat	Specify the central latitude of the range to be acquired, decimal notation, and geodetic system is WGS84	☐
lon	Specify the central longitude of the range to be acquired, decimal notation, and geodetic system is WGS84	☐
radius	Specify the radius of the area to be acquired in meters, in the range of 0 - 4,000 meters	☐
acl:consumerKey	Specify an access token issued on the developer site.	☐
PREDICATE	Filter by specifying the property of the className specified by rdf:type	

<https://developer.odpt.org/documents>

PREDICATE is a property name that changes for each rdf:type.

Request example

```
curl -X GET 'https://api.odpt.org/api/v4/places/odpt:Station?lon=139.766926&lat=35.681265&ra'
```

1.7.1. Filtering

By specifying PREDICATE as a parameter, you can filter search results by a specific value.

Example request: dc:title=東京

```
curl -X GET 'https://api.odpt.org/api/v4/places/odpt:Station?lon=139.766926&lat=35.681265&ra'
```

When specifying multiple search conditions, "OR" search is requested by specifying the value of PREDICATE separated by a comma ",".

Example request: dc:title=東京,大手町

```
curl -X GET 'https://api.odpt.org/api/v4/places/odpt:Station?lon=139.766926&lat=35.681265&ra'
```

2. Common className

2.1. Overview

The common className defines the following data common to various data types of public transportation.

- odpt:Calendar
- odpt:Operator

For the contents of each data, refer to each explanation.

2.1.1. Version information

Version : 4.0.0

2.1.2. URI scheme

Host : api.odpt.org

BasePath : /api/v4

Schemes : HTTPS

2.1.3. MIME type

- application/json

2.2. Paths

2.2.1. GET /api/v4/odpt:Calendar

Description

Acquire the contents of day of the week / date category (odpt:Calendar).

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)

Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
-------	-------------------------------	---	--------------

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:Calendar > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:Calendar?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:ucode:_00001C0000000000000010000030FD7E5",
  "@type" : "odpt:Calendar",
  "dc:date" : "2017-01-13T15:10:00+09:00",
  "owl:sameAs" : "odpt.Calendar:Weekday",
  "dc:title" : "平日",
  "odpt:calendarTitle" : {
    "ja" : "平日",
    "en" : "Weekday"
  }
} ]
```

2.2.2. GET /api/v4/odpt:Operator

Description

Acquire the contents of the public transportation operator (odpt:Operator).

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:Operator > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:Operator?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:ucode:_00001C000000000000000100000030FD7E5",
  "@type" : "odpt:Operator",
  "dc:date" : "2017-01-13T15:10:00+09:00",
  "owl:sameAs" : "odpt.Operator:JR-East",
  "dc:title" : "JR東日本",
  "odpt:operatorTitle" : {
    "ja" : "JR東日本",
    "en" : "JR East"
  }
} ]
```

2.3. Definitions

2.3.1. odpt:Calendar

ClassName that handles day of the week / date classNameification. Holds a list of target dates.

Base className

The odpt:Calendar base className is a className for expressing general date classNameifications. Each exists as an instance of type odpt:Calendar.

- odpt.Calendar:Weekday: Weekdays (Monday to Friday, excluding holidays)
- odpt.Calendar:Holiday: Holiday (Sunday, holiday, off day, substitute holiday)
- odpt.Calendar:SaturdayHoliday: Saturday / off day (Saturday or off day)
- odpt.Calendar:Sunday: Sunday
- odpt.Calendar:Monday: Monday
- odpt.Calendar:Tuesday: Tuesday

- odpt.Calendar:Wednesday: Wednesday
- odpt.Calendar:Thursday: Thursday
- odpt.Calendar:Friday: Friday
- odpt.Calendar:Saturday: Saturday

Specific className

- odpt.Calendar:Specific.Foo.Bar
 - E.g. odpt.Calendar:Specific.Toei.MarketHoliday: market holiday

If more than one calendar applies to a given date, the priorities are as follows:

- Specific classNamees (Specific.XXX) take precedence over base classNamees.
- If the base className holiday (Holiday) overlaps with Saturday (Saturday), the holiday takes precedence.
- If more than one specific classNamees are applicable, the timetable for that date is a merge of the timetables for these specific classNamees.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	ClassName name. odpt.Calendar is returned here. Example : <code>"odpt:Calendar"</code>	string
dc:date <i>optional</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.Calendar:name(.period). Example : <code>"odpt.Calendar:Weekday"</code>	string (URL)
dc:title <i>optional</i>	Calendar name (Japanese). E.g. 平日, 休日, 市場休日 Example : <code>"平日"</code>	string
odpt:calendarTitle <i>optional</i>	Calendar name (multilingual-support) Example : <code>{"ja" : "平日", "en" : "Weekday"}</code>	object
odpt:day <i>optional</i>	A list of applicable dates. All values are in ISO8601 date format. Example : <code>["2017-01-13", "2017-01-18"]</code>	< string (xsd:date) > array
odpt:duration <i>optional</i>	Calendar validity period. All values are in ISO8601 period format. Example : <code>"2017-11-13/2017-11-18"</code>	string

2.3.2. odpt:Operator

ClassName dealing with public transportation operators

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	ClassName name. odpt.Operator is returned here. Example : <code>"odpt:Operator"</code>	string

dc:date <i>optional</i>	Data generation date and time (ISO8601 date and time format) Example : "2017-01-13T15:10:00+09:00"	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.Operator:name Example : "odpt.Operator:XX"	string (URL)
dc:title <i>optional</i>	Operator name (Japanese) Example : "〇〇電鉄"	string
odpt:operatorTitle <i>optional</i>	Operator name (multilingual-support) Example : {"ja" : "〇〇電鉄", "en" : "XX Railway"}	object

3. ODPT Train API

3.1. Overview

With Train API, the following data related to railway can be acquired.

- odpt:PassengerSurvey
- odpt:Railway
- odpt:RailDirection
- odpt:RailwayFare
- odpt:Station
- odpt:StationTimetable
- odpt:Train
- odpt:TrainInformation
- odpt:TrainTimetable
- odpt:TrainType

For the contents of each data, refer to each explanation.

3.1.1. Version information

Version : 4.0.0

3.1.2. URI scheme

Host : api.odpt.org

BasePath : /api/v4

Schemes : HTTPS

3.1.3. MIME type

- application/json

3.2. Paths

3.2.1. GET /api/v4/odpt:PassengerSurvey

Description

Acquire the number of people getting on and off at a station.

Parameters

Type	Name	Description	Schema
------	------	-------------	--------

Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:operator <i>optional</i>	ID representing the operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:station <i>optional</i>	ID representing the station (owl:sameAs of odpt:Station)	string (odpt:Station owl:sameAs)
Query	odpt:railway <i>optional</i>	ID representing the line (owl:sameAs of odpt:Railway)	string (odpt:Railway owl:sameAs)

HTTP Code	Description	Schema
200	Successful completion	< odpt:PassengerSurvey > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Request path

```
/api/v4/odpt:PassengerSurvey?acl:consumerKey=string
```

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:ucode:_00001C000000000000010000030FD7E5",
  "@type" : "odpt:PassengerSurvey",
  "dc:date" : "2017-01-13T06:10:00+0000",
  "owl:sameAs" : "odpt.PassengerSurvey:JR-East.Tokyo",
  "odpt:operator" : "odpt.Operator:JR-East",
  "odpt:railway" : [ "odpt.Railway:JR-East.Yamanote", "odpt.Railway:JR-E"
  "odpt:station" : [ "odpt.Station:JR-East.Yamanote.Tokyo", "odpt.Static
  "odpt:includeAlighting" : false,
  "odpt:passengerSurveyObject" : [ {
    "odpt:surveyYear" : 2016,
    "odpt:passengerJourneys" : 12340
  }, {
```

```
"odpt:surveyYear" : 2017,
"odpt:passengerJourneys" : 12345
} ]
```

Description

Acquire the definition of the direction of travel.

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:RailDirection > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:RailDirection?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
    "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
    "@id" : "urn:ucode:_00001C0000000000000010000030FD7E5",
    "@type" : "odpt:RailDirection",
    "dc:date" : "2017-01-13T06:10:00+0000",
    "owl:sameAs" : "odpt.RailDirection:Inbound",
    "dc:title" : "上り",
    "odpt:railDirectionTitle" : {
        "ja" : "上り",
        "en" : "Inbound"
    }
}
```

Description

Acquire line information.

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	dc:title <i>optional</i>	Line name (e.g. 小田原線,京王線, 相模原線)	string
Query	odpt:operator <i>optional</i>	ID representing the operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:lineCode <i>optional</i>	Line code, line symbol notation (e.g. Odawara Line=> OH, Marunouchi Line=> M)	string

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:Railway > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:Railway?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
    "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
    "@id" : "urn:uicode:_00001C0000000000000010000030FD7E5",
    "@type" : "odpt:Railway",
```



```
    "dc:date" : "2017-01-13T15:10:00+09:00",
    "owl:sameAs" : "odpt:Railway:JR-East.ChuoRapid",
    "dc:title" : "中央線快速",
    "odpt:railwayTitle" : {
      "ja" : "中央線快速",
      "en" : "Chuo Rapid Line"
    },
    "odpt:operator" : "odpt.Operator:JR-East",
    "odpt:ascendingRailDirection" : "odpt.RailDirection:Outbound",
    "odpt:descendingRailDirection" : "odpt.RailDirection:Inbound",
    "odpt:stationOrder" : [ {
      "odpt:index" : 1,
      "odpt:station" : "odpt.Station:JR-East.ChuoRapid.Tokyo",
      "odpt:stationTitle" : {
        "ja" : "東京",
        "en" : "Tokyo"
      }
    }, {
      "odpt:index" : 2,
      "odpt:station" : "odpt.Station:JR-East.ChuoRapid.Kanda",
      "odpt:stationTitle" : {
        "ja" : "神田",
        "en" : "Kanda"
      }
    }, {
      "odpt:index" : 3,
      "odpt:station" : "odpt.Station:JR-East.ChuoRapid.Ochanomizu",
      "odpt:stationTitle" : {
        "ja" : "御茶ノ水",
        "en" : "Ochanomizu"
      }
    }
  ]
}
```

3.2.4. GET /api/v4/odpt:RailwayFare

Description

Acquire fare between two stations.

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:operator <i>optional</i>	ID representing the operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:fromStation <i>optional</i>	ID of the starting station (owl:sameAs of odpt:Station)	string (odpt:Station owl:sameAs)
Query	odpt:toStation <i>optional</i>	ID of destination station (owl:sameAs of odpt:Station)	string (odpt:Station owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:RailwayFare > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:RailwayFare?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:ucode:_00001C00000000000000010000030FD7E5",
  "@type" : "odpt:RailwayFare",
  "dc:date" : "2017-01-13T15:10:00+09:00",
  "owl:sameAs" : "odpt.RailwayFare:TokyoMetro.Marunouchi.Tokyo.TokyoMetro",
  "odpt:operator" : "odpt.Operator:TokyoMetro",
  "odpt:fromStation" : "odpt.Station:TokyoMetro.Marunouchi.Tokyo",
  "odpt:toStation" : "odpt.Station:TokyoMetro.Tozai.Nakano",
  "odpt:ticketFare" : 240,
  "odpt:icCardFare" : 237,
  "odpt:childTicketFare" : 120,
}
```

3.2.5. GET /api/v4/odpt:Station

Description

Acquire station information.

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	dc:title <i>optional</i>	Station name (e.g. 東京, 新宿, and 上野)	string

Query	odpt:operator <i>optional</i>	ID representing the operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:railway <i>optional</i>	Line ID where the station is located (owl:sameAs of odpt:Railway)	string (odpt:Railway owl:sameAs)
Query	odpt:stationCode <i>optional</i>	Station numbering (e.g. OH01=Odakyu Shinjuku Station)	string

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:Station > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:Station?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:ucode:_00001C000000000000000100000030FD7E5",
  "@type" : "odpt:Station",
  "dc:date" : "2017-01-13T15:10:00+09:00",
  "owl:sameAs" : "odpt.Station:JR-East.Yamanote.Tokyo",
  "dc:title": "東京",
  "odpt:stationTitle" : {
    "ja" : "東京",
    "en" : "Tokyo"
  },
  "odpt:operator" : "odpt.Operator:JR-East",
  "odpt:railway" : "odpt.Railway:JR-East.Yamanote",
  "geo:long" : 139.1234,
  "geo:lat" : 35.1234
} ]
```

3.2.6. GET /api/v4/odpt:StationTimetable

Description

Acquire the station timetable.

Parameters

--	--	--	--

Responses

Example HTTP request

```
/api/v4/odpt:StationTimetable?acl:consumerKey=string
```

```
[ {
    "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
    "@id" : "urn:ucode: 00001C0000000000000010000030FD7E5",
    "label" : "00001C0000000000000010000030FD7E5",
    "type" : "ucode"
}]
```

```
"@type" : "odpt:StationTimetable",
"dc:date" : "2017-01-13T15:10:00+09:00",
"dct:issued" : "2017-01-01",
"owl:sameAs" : "odpt.StationTimetable:JR-East.ChuoRapid.Tokyo.Outbound",
"odpt:operator" : "odpt.Operator:JR-East",
"odpt:railway" : "odpt.Railway:JR-East.ChuoRapid",
"odpt:station" : "odpt.Station:JR-East.ChuoRapid.Tokyo",
"odpt:railDirection" : "odpt.RailDirection:JR-East.Outbound",
"odpt:calendar" : "odpt.Calendar:Weekday",
"odpt:stationTimetableObject" : [ {
  "odpt:departureTime" : "06:00",
  "odpt:destinationStation" : [ "odpt.Station:JR-East.ChuoRapid.Takao" ]
}, {
  "odpt:departureTime" : "06:10",
  "odpt:destinationStation" : [ "odpt.Station:JR-East.ChuoRapid.Takao" ]
}, {
  "odpt:departureTime" : "06:20",
  "odpt:destinationStation" : [ "odpt.Station:JR-East.ChuoRapid.Takao" ]
} ]
}
```

3.2.7. GET /api/v4/odpt:Train

Description

Acquire train information (position information of train).

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:operator <i>optional</i>	ID of the operator that dispatches train information (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:railway <i>optional</i>	ID of the line serviced by this train (owl:sameAs of odpt:Railway)	string (odpt:Railway owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:Train > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

--	--	--

Example HTTP request

Request path

```
/api/v4/odpt:Train?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:uuid:5d7dd592-ef17-4b69-b955-5b4fe5f7e350",
  "@type" : "odpt:Train",
  "dc:date" : "2017-12-07T01:24:33+09:00",
  "owl:sameAs" : "odpt.Train:JR-East.Utsunomiya.565M",
  "odpt:operator" : "odpt.Operator:JR-East",
  "odpt:railway" : "odpt.Railway:JR-East.Utsunomiya",
  "odpt:railDirection" : "odpt.RailwayDirection:Outbound",
  "odpt:trainNumber" : "565M",
  "odpt:fromStation" : "odpt.Station:JR-East.Utsunomiya.Suzumenomiya",
  "odpt:toStation" : "odpt.Station:JR-East.Utsunomiya.Utsunomiya",
  "odpt:destinationStation" : [ "odpt.Station:JR-East.Utsunomiya.Utsunomiya",
  "odpt:index" : 1,
  "odpt:delay" : 0,
  "odpt:carComposition" : 15
} ]
```

3.2.8. GET /api/v4/odpt:TrainInformation

Description

Acquire train operation information.

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:operator <i>optional</i>	ID of the operator that dispatches operation information (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:railway <i>optional</i>	ID of the line where the operation information occurred (owl:sameAs of odpt:Railway)	string (odpt:Railway owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:TrainInformation > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content

404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:TrainInformation?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
    "@context" : "http://vocab.odpt.org/context_odpt.json",
    "@id" : "urn:ucode:_00001C0000000000000010000030C3BE4",
    "@type" : "odpt:TrainInformation",
    "dc:date" : "2017-12-07T01:25:03+09:00",
    "owl:sameAs" : "odpt.TrainInformation:TokyoMetro.Ginza",
    "dct:valid" : "2017-12-07T01:30:03+09:00",
    "odpt:timeOfOrigin" : "2017-11-21T11:31:00+09:00",
    "odpt:operator" : "odpt.Operator:TokyoMetro",
    "odpt:railway" : "odpt.Railway:TokyoMetro.Ginza",
    "odpt:trainInformationText" : {
        "ja" : "現在、平常どおり運転しています。",
        "en" : "Running on schedule."
    }
} ]
```

3.2.9. GET /api/v4/odpt:TrainTimetable

Description

Acquire train timetable.

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier	string (URN)
Query	owl:sameAs <i>optional</i>	Unique identifier alias	string (URL)
Query	odpt:trainNumber <i>optional</i>	Train number	string
Query	odpt:railway <i>optional</i>	Line ID (owl:sameAs of odpt:Railway)	string (odpt:Railway owl:sameAs)
Query	odpt:operator <i>optional</i>	ID of operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:trainType <i>optional</i>	Train type ID (owl:sameAs of odpt:TrainType)	string (odpt:TrainType owl:sameAs)

Query	odpt:train <i>optional</i>	ID used for locating the train (owl:sameAs of odpt:Train)	string (odpt:Train owl:sameAs)
Query	odpt:calendar <i>optional</i>	Specific calendar information ID (owl:sameAs of odpt:Calendar)	string (odpt:Calendar owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful response	< odpt:TrainTimetable > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:TrainTimetable?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {  
    "@context" : "http://vocab.odpt.org/context_odpt.jsonld",  
    "@id" : "urn:ucode:_00001C0000000000000010000030FD7E5",  
    "@type" : "odpt:TrainTimetable",  
    "dc:date" : "2017-01-13T15:10:00+09:00",  
    "dct:issued" : "2017-01-01",  
    "owl:sameAs" : "odpt.TrainTimetable:JR-East.ChuoRapid.123M.Weekday",  
    "odpt:operator" : "odpt.Operator:JR-East",  
    "odpt:railway" : "odpt.Railway:JR-East.ChuoRapid",  
    "odpt:railDirection" : "odpt.RailDirection:Outbound",  
    "odpt:calendar" : "odpt.Calendar:Weekday",  
    "odpt:trainNumber" : "123M",  
    "odpt:trainType" : "odpt.TrainType:JR-East.Rapid",  
    "odpt:originStation" : [ "odpt.Station:JR-East.ChuoRapid.Tokyo" ],  
    "odpt:destinationStation" : [ "odpt.Station:JR-East.ChuoRapid.Takao" ]  
    "odpt:trainTimetableObject" : [ {  
        "odpt:departureTime" : "06:00",  
        "odpt:departureStation" : "odpt.Station:JR-East.ChuoRapid.Tokyo"  
    } , {  
        "odpt:departureTime" : "06:30",  
        "odpt:departureStation" : "odpt.Station:JR-East.ChuoRapid.Tachikawa"  
    } , {  
        "odpt:arrivalTime" : "07:00",  
        "odpt:arrivalStation" : "odpt.Station:JR-East.ChuoRapid.Takao"  
    } ]  
} ]
```


Description

Parameters

Responses

Example HTTP request

```
/api/v4/odpt:TrainType?acl:consumerKey=string
```

Response 200

```
[ {
    "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
    "@id" : "urn:ucode:_00001C000000000000010000030FD7E5",
    "@type" : "odpt:TrainType",
    "dc:date" : "2017-01-13T15:10:00+09:00",
    "owl:sameAs" : "odpt.TrainType:JR-East.Local",
    "odpt:operator" : "odpt.Operator:JR-East",
    "dc:title" : "普通",
    "odpt:trainTypeTitle" : {
        "ja" : "普通",
        "en" : "Local"
    }
} ]
```

3.3. Definitions

3.3.1. odpt:PassengerSurvey

Shows the number of people getting on and off at a station or the number of people getting on. The data is the value published by each operator.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	ClassName name. odpt:PassengerSurvey Example : <code>"odpt:PassengerSurvey"</code>	string
dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.PassengerSurvey:operatorname.stationname or odpt.PassengerSurvey:operatorname.linename.stationname. Example : <code>"odpt.PassengerSurvey:JR-East.Tokyo"</code>	string (URL)
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:station <i>required</i>	A list of IDs (owl:sameAs of odpt:Station) representing stations Example : <code>["odpt.Station:JR-East.Yamanote.Tokyo", "odpt.Station:JR-East.ChuoRapid.Tokyo"]</code>	< string (odpt:Station owl:sameAs) > array
odpt:railway <i>required</i>	A list of IDs (owl:sameAs of odpt:Railway) representing lines Example : <code>["odpt.Railway:JR-East.Yamanote", "odpt.Railway:JR-East.ChuoRapid"]</code>	< string (odpt:Railway owl:sameAs) > array
odpt:includeAlighting <i>required</i>	True for requesting the number of passengers getting on and off, false for passengers getting on but not getting off Example : <code>true</code>	boolean (xsd:boolean)
odpt:passengerSurveyObject <i>required</i>	A list of pairs of the survey year and the average number of passengers getting on and off (or the number of passengers getting on) Example : <code>[{"odpt:surveyYear" : 2017, "odpt:passengerJourneys" : 12345}]</code>	< odpt:PassengerSurveyObject > array

odpt:PassengerSurveyObject

Name	Description	Schema
odpt:surveyYear <i>required</i>	Survey year Example : <code>2017</code>	integer
odpt:passengerJourneys <i>required</i>	Average number of passengers getting on and off at the station per day (or number of passengers getting on) Example : <code>12345</code>	integer

3.3.2. odpt:RailDirection

Defines the direction of travel of the train, such as going up or down.

Depending on the railway company or line, even when direction is given as southbound or northbound, it may be represented by odpt.RailDirection:Inbound or odpt.RailDirection:Outbound for convenience.

Name	Description	Schema
@context <i>required</i>	@Context URL based on JSON-LD specification Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:RailDirection Example : <code>"odpt:RailDirection"</code>	string
dc:date <i>optional</i>	Generation date of direction information (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.RailDirection:direction or odpt.RailDirection:operatorname.direction. Example : <code>"odpt.RailDirection:Inbound"</code>	string (URL)
dc:title <i>optional</i>	Direction of travel (Japanese) Example : <code>"上り"</code>	string
odpt:railDirectionTitle <i>optional</i>	Direction of travel (multilingual-support) Example : <code>{"ja" : "上り", "en" : "Inbound"}</code>	object

3.3.3. odpt:Railway

Shows information about railway lines (operating routes). The operating route name dc:title represents a commonly used line name and nickname. For example, if dc:title is 山手線, it represents information on the 山手線, which is generally recognized in Tokyo.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:Railway Example : <code>"odpt:Railway"</code>	string
dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.Railway:operatorname.linename. Example : <code>"odpt.Railway:JR-East.Yamanote"</code>	string (URL)
dc:title <i>required</i>	Line name (Japanese) Example : <code>"山手線"</code>	string
odpt:railwayTitle <i>optional</i>	Line name (multilingual-support) Example : <code>{"ja" : "山手線", "en" : "Yamanote Line"}</code>	object
odpt:kana	Line name Yomigana (Hiragana notation)	string

<i>optional</i>	Example : <code>"string"</code>	
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:lineCode <i>optional</i>	Stores line code and line symbol notation. E.g. 丸ノ内線=>M Example : <code>"M"</code>	string
odpt:color <i>optional</i>	Expresses the line color of the line in HEX notation #RRGGBB, DIC notation DICnnn, PANTONE notation PANTONExxxx Example : <code>"string"</code>	string
ug:region <i>optional</i>	Feature information in GeoJSON format Example : <code>"object"</code>	object
odpt:ascendingRailDirection <i>optional</i>	ID representing the ascending direction. Stores the ascending direction of odpt:index of odpt:stationOrder by ID represented with owl:sameAs of odpt:RailDirection. Example : <code>"odpt.RailDirection:Outbound"</code>	string (odpt:RailDirection owl:sameAs)
odpt:descendingRailDirection <i>optional</i>	ID representing the descending direction. Stores the descending direction of odpt:index of odpt:stationOrder with the ID represented by owl:sameAs of odpt:RailDirection. Example : <code>"odpt.RailDirection:Inbound"</code>	string (odpt:RailDirection owl:sameAs)
odpt:stationOrder <i>required</i>	A list representing station order Example : <code>[{ "odpt:index" : 1, "odpt:station" : "odpt.Station:JR-East.ChuoRapid.Tokyo" }]</code>	< odpt:StationOrder > array

odpt:StationOrder

Name	Description	Schema
odpt:station <i>required</i>	ID representing the station (owl:sameAs of odpt:Station) Example : <code>"odpt.Station:TokyoMetro.Marunouchi.Tokyo"</code>	string (odpt:Station owl:sameAs)
odpt:stationTitle <i>optional</i>	Station name (multilingual-support) Example : <code>{ "ja" : "東京", "en" : "Tokyo" }</code>	object
odpt:index <i>required</i>	Integer value representing station order. In principle, the train stops in ascending or descending order of this value, depending on the direction of travel. In the case of a loop line, the same station may be described more than once. Example : <code>1</code>	integer

3.3.4. odpt:RailwayFare

Shows fare information between two stations. This information is constructed based on the information provided by each transportation operator.

Name	Description
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C000000000000010000030FD7E5"</code>
@type <i>required</i>	Name of the className. odpt:RailwayFare Example : <code>"odpt:RailwayFare"</code>

dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>
dct:issued <i>optional</i>	Fare revision date (ISO8601 date format) Example : <code>"2017-01-13"</code>
dct:valid <i>optional</i>	Data guarantee period (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.RailwayFare:departurestation.operatorname.departurestationlinename.departurestationname.arrivalstation.operatorname.arriv Example : <code>"odpt.RailwayFare:TokyoMetro.Marunouchi.Tokyo.TokyoMetro.Tozai.Nakano"</code>
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:TokyoMetro"</code>
odpt:fromStation <i>required</i>	ID of the starting station (owl:sameAs of odpt:Station) Example : <code>"odpt.Station:TokyoMetro.Marunouchi.Tokyo"</code>
odpt:toStation <i>required</i>	ID of destination station (owl:sameAs of odpt:Station) Example : <code>"odpt.Station:TokyoMetro.Tozai.Nakano"</code>
odpt:ticketFare <i>required</i>	Fare for ticket Example : <code>200</code>
odpt:icCardFare <i>optional</i>	Fare for IC card Example : <code>196</code>
odpt:childTicketFare <i>optional</i>	Child fare for ticket Example : <code>100</code>
odpt:childIcCardFare <i>optional</i>	Child fare for IC card Example : <code>98</code>
odpt:viaStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing transit stations for the fare calculation Example : <code>["odpt.Station:TokyoMetro.Tozai.NishiFunabashi"]</code>
odpt:viaRailway <i>optional</i>	A list of IDs (owl:sameAs of odpt:Railway) representing transit lines for the fare calculation Example : <code>["odpt.Railway:TokyoMetro.Tozai"]</code>
odpt:ticketType <i>optional</i>	Ticket type (If the fare differs depending on the train type, such as limited express or liner) Example : <code>"string"</code>
odpt:paymentMethod <i>optional</i>	A list of payment methods Example : <code>["string"]</code>

3.3.5. odpt:Station

Represents station information. This information is constructed based on the information provided by each transportation operator.

Name	Description	Schema
@context <i>required</i>	@Context URL based on JSON-LD specification	string (URL)

	Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:Station Example : <code>"odpt:Station"</code>	string
dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.Station:operatorname.linename.stationname. Example : <code>"odpt.Station:JR-East.Yamanote.Tokyo"</code>	string (URL)
dc:title <i>optional</i>	Station name (Japanese) Example : <code>"東京"</code>	string
odpt:stationTitle <i>optional</i>	Station name (multilingual-support) Example : <code>{"ja" : "東京", "en" : "Tokyo"}</code>	object
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:railway <i>required</i>	ID representing the line (owl:sameAs of odpt:Railway) Example : <code>"odpt.Railway:JR-East.Yamanote"</code>	string (odpt:Railway owl:sameAs)
odpt:stationCode <i>optional</i>	Station code Example : <code>"JY01"</code>	string
geo:long <i>optional</i>	Longitude of representative point (decimal notation, geodetic system is WGS84) Example : <code>139.1234</code>	number
geo:lat <i>optional</i>	Latitude of representative point (decimal notation, geodetic system is WGS84) Example : <code>35.1234</code>	number
ug:region <i>optional</i>	Feature information in GeoJSON format Example : <code>"object"</code>	object
odpt:exit <i>optional</i>	A list of IDs representing station entrances. For the ID, use the value of @id of ug:Poi. Example : <code>["string"]</code>	< string > array
odpt:connectingRailway <i>optional</i>	A list of IDs of transferable lines (owl:sameAs in odpt:Railway) Example : <code>["odpt.Railway:JR-East.ChuoRapid", "odpt.Railway:TokyoMetro.Marunouchi"]</code>	< string (odpt:Railway owl:sameAs) > array
odpt:connectingStation <i>optional</i>	A list of IDs of transferable stations (owl:sameAs in odpt:Station) Example : <code>["odpt.Station:JR-East.ChuoRapid.Tokyo", "odpt.Station:TokyoMetro.Marunouchi.Otemachi"]</code>	< string (odpt:Station owl:sameAs) > array
odpt:stationTimetable <i>optional</i>	A list of IDs (owl:sameAs of odpt:StationTimetable) representing the station timetables Example : <code>["odpt.StationTimeTable:JR-East.Yamanote.Tokyo.OuterLoop.Weekday"]</code>	< string (odpt:StationTimetable owl:sameAs) > array
odpt:passengerSurvey <i>optional</i>	A list of IDs (owl:sameAs of odpt:PassengerSurvey) representing the number of people getting on and off at the station Example : <code>["odpt.PassengerSurvey:JR-East.Tokyo"]</code>	< string (odpt:PassengerSurvey owl:sameAs) > array

3.3.6. odpt:StationTimetable

Station timetable information classNameified for each station, line, and direction is shown.

- If the midnight comes during transfer, odpt:arrivalTime and odpt:departureTime are expressed in time from 00:00 to 23:59, so the departure time between station A and station B is 23:58 at station A -> station B at 00:03, for example. Therefore, in order to determine if the day rolls over, it is necessary for the client to determine that the day has rolled over when the time (hour) changes from the previous station's from 23 to 0.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:StationTimetable Example : <code>"odpt:StationTimetable"</code>	string
dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dct:issued <i>optional</i>	Schedule revision date (ISO8601 date format) Example : <code>"2017-01-13"</code>	string (xsd:date)
dct:valid <i>optional</i>	Data guarantee period (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.StationTimetable:operatorname.linename.stationname.directionname.thedayoftheweektype. Example : <code>"odpt.StationTimetable:JR-East.ChuoRapid.Tokyo.Outbound.Weekday"</code>	string (URL)
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:railway <i>required</i>	ID representing the line (owl:sameAs of odpt:Railway) Example : <code>"odpt.Railway:JR-East.ChuoRapid"</code>	string (odpt:Railway owl:sameAs)
odpt:railwayTitle <i>optional</i>	Line name (multilingual-support) Example : <code>{"ja" : "中央線快速", "en" : "Chuo Rapid"}</code>	object
odpt:station <i>optional</i>	ID representing the station (owl:sameAs of odpt:Station) Example : <code>"odpt.Station:JR-East.ChuoRapid.Tokyo"</code>	string (odpt:Station owl:sameAs)
odpt:stationTitle <i>optional</i>	Station name (multilingual-support) Example : <code>{"ja" : "東京", "en" : "Tokyo"}</code>	object
odpt:railDirection <i>optional</i>	ID representing the direction (owl:sameAs of odpt:RailDirection) Example : <code>"odpt.RailDirection:Outbound"</code>	string (odpt:RailDirectio owl:sameAs)
odpt:calendar <i>optional</i>	ID of the day of the week / date information for the operation (owl:sameAs of odpt:Calendar) Example : <code>"odpt.Calendar:Weekday"</code>	string (odpt:Calendar owl:sameAs)
odpt:stationTimetableObject <i>required</i>	A list of pairs of departure time, final (destination) station, etc. Example : <code>[{"odpt:departureTime" : "06:00"}, {"odpt:departureTime" : "07:00"}]</code>	< odpt:stationTimetableC > array
odpt:note <i>optional</i>	Natural language description of annotation information not defined as properties (multilingual-support) Example : <code>{"ja" : "日本語での注釈情報", "en" : "Note in English"}</code>	object

odpt:stationTimetableObject

Name	Description	Schema
odpt:arrivalTime <i>optional</i>	Arrival time (ISO8601 time format) Example : <code>"05:08"</code>	string (odpt:Time)
odpt:departureTime <i>optional</i>	Departure time (ISO8601 time format) Example : <code>"05:09"</code>	string (odpt:Time)
odpt:originStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the departing train stations Example : <code>["odpt:Station:JR-East.ChuoRapid.Tokyo"]</code>	< string (odpt:Station owl:sameAs) > array
odpt:destinationStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the final stations (destination stations) Example : <code>["odpt:Station:JR-East.ChuoRapid.Takao"]</code>	< string (odpt:Station owl:sameAs) > array
odpt:viaStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the transfer stations Example : <code>["odpt:Station:TokyoMetro.Tozai.NishiFunabashi"]</code>	< string (odpt:Station owl:sameAs) > array
odpt:viaRailway <i>optional</i>	A list of IDs (owl:sameAs of odpt:Railway) representing transfer lines Example : <code>["odpt:Railway:TokyoMetro.Tozai"]</code>	< string (odpt:Railway owl:sameAs) > array
odpt:train <i>optional</i>	ID of train that arrives or departs (owl:sameAs of odpt:Train) Example : <code>"odpt:Train:JR-East.Yamanote.123M"</code>	string (odpt:Train owl:sameAs)
odpt:trainNumber <i>optional</i>	Train number Example : <code>"123M"</code>	string
odpt:trainType <i>optional</i>	Train type ID (owl:sameAs of odpt:TrainType) Example : <code>"odpt:TrainType:JR-East.Local"</code>	string (odpt:TrainType owl:sameAs)
odpt:trainName <i>optional</i>	A list of train service names and nicknames (multilingual-support) Example : <code>[{ "ja" : "むさし", "en" : "Musashi" }]</code>	< object > array
odpt:trainOwner <i>optional</i>	ID representing the operator that owns the vehicle (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:isLast <i>optional</i>	True if the last train. Omitted if not the last train. Example : <code>true</code>	boolean (xsd:boolean)
odpt:isOrigin <i>optional</i>	True if the first train station. Omitted if not the first station. Example : <code>true</code>	boolean (xsd:boolean)
odpt:platformNumber <i>optional</i>	Platform number where the train arrives Example : <code>"1"</code>	string
odpt:platformName <i>optional</i>	The name of the platform where the train arrives or departs (multilingual-support) Example : <code>{ "ja" : "日本語名称", "en" : "English Name" }</code>	object
odpt:carComposition <i>optional</i>	Number of cars (stored when the number of cars that stop at the station differs for different trains) Example : <code>8</code>	integer
odpt:note <i>optional</i>	Natural language description of annotation information not defined as properties (multilingual-support) Example : <code>{ "ja" : "日本語での注釈情報", "en" : "Note in English" }</code>	object

3.3.7. odpt:Train

Shows the real-time position information of a train. Specifically, the relative position information is described using `odpt:fromStation`, and `odpt:toStation`, i.e., "moving from a station to another station."

Current location	odpt:fromStation	odpt:toStation
Located between station A and station B	Station A	Station B
Located near station A	Station A	null

Being located means either a case where the train is traveling (passing) or a case where the train is stopped.

If the information is not available regarding whether a train is either travelling or stopped near station A, `odpt:toStation` may not be null.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode or uuid) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. <code>odpt:Train</code> Example : <code>"odpt:Train"</code>	string
dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dct:valid <i>optional</i>	Data guarantee period (ISO8601 date and time format) Example : <code>"2017-01-13T15:15:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is <code>odpt.Train:operatorname.linename.trainnumber</code> . Example : <code>"odpt.Train:JR-East.Yamanote.123M"</code>	string (URL)
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of <code>odpt:Operator</code>) Example : <code>"odpt.Operator:JR-East"</code>	string (<code>odpt:Operator</code> owl:sameAs)
odpt:railway <i>required</i>	ID representing the railway line (odpt:Railway's owl:sameAs) Example : <code>"odpt.Railway:JR-East.Yamanote"</code>	string (<code>odpt:Railway</code> owl:sameAs)
odpt:railDirection <i>optional</i>	ID representing the direction of travel (owl:sameAs of <code>odpt:RailDirection</code>) Example : <code>"odpt.RailDirection:Outbound"</code>	string (<code>odpt:RailDirection</code> owl:sameAs)
odpt:trainNumber <i>required</i>	Train number Example : <code>"123M"</code>	string
odpt:trainType <i>optional</i>	Train type (owl:sameAs of <code>odpt:TrainType</code>) Example : <code>"odpt.TrainType:JR-East.Local"</code>	string (<code>odpt:TrainType</code> owl:sameAs)
odpt:trainName <i>optional</i>	A list of train service names and nicknames (multilingual-support) Example : <code>[{ "ja" : "むさし", "en" : "Musashi" }]</code>	< object > array
	Stores the station ID (owl:sameAs of <code>odpt:Station</code>) representing the station which	string

odpt:fromStation <i>optional</i>	the train just left or where it has stopped. See the beginning of the description of odpt:Train for details. Example : <code>"odpt.Station:JR-East.ChuoRapid.Tokyo"</code>	(odpt:Station owl:sameAs)
odpt:toStation <i>optional</i>	Stores the ID (owl:sameAs of odpt:Station) representing the station to which the train is heading. See the beginning of the description of odpt:Train for details. Example : <code>"odpt.Station:JR-East.ChuoRapid.Tokyo"</code>	string (odpt:Station owl:sameAs)
odpt:originStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the starting station of the train service Example : <code>["odpt.Station:JR-East.ChuoRapid.Tokyo"]</code>	< string (odpt:Station owl:sameAs) > array
odpt:destinationStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the final station (destination station) of the train service Example : <code>["odpt.Station:JR-East.ChuoRapid.Takao"]</code>	< string (odpt:Station owl:sameAs) > array
odpt:viaStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the transfer-stations of the train service Example : <code>["odpt.Station:TokyoMetro.Tozai.NishiFunabashi"]</code>	< string (odpt:Station owl:sameAs) > array
odpt:viaRailway <i>optional</i>	A list of IDs (owl:sameAs of odpt:Railway) representing the transfer lines where the train passes Example : <code>["odpt.Railway:TokyoMetro.Tozai"]</code>	< string (odpt:Railway owl:sameAs) > array
odpt:trainOwner <i>optional</i>	ID representing the operator that owns the vehicle (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:index <i>optional</i>	Train order within a line Example : <code>1</code>	integer
odpt:delay <i>optional</i>	Delay time (seconds) Example : <code>60</code>	integer
odpt:carComposition <i>optional</i>	Number of cars Example : <code>8</code>	integer
odpt:note <i>optional</i>	Natural language description of annotation information not defined as properties (multilingual-support) Example : <code>{"ja" : "日本語での注釈情報", "en" : "Note in English"}</code>	object

3.3.8. odpt:TrainInformation

Represents real-time train operation information. There are lines where operation information is generated only when an operation failure occurs, and lines where a character string including a character string such as "normal" is returned even during normal operation.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode or uuid) Example : <code>"urn:ucode:_00001C000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:TrainInformation Example : <code>"odpt:TrainInformation"</code>	string
dc:date	Data generation date and time (ISO8601 date and time format)	string

<i>required</i>	Example : <code>"2017-01-13T15:10:00+09:00"</code>	(xsd:dateTime)
dct:valid <i>optional</i>	Data guarantee period (ISO8601 date and time format) Example : <code>"2017-01-13T15:15:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt:TrainInformation:operatorname.linename or odpt:TrainInformation:operatorname. Example : <code>"odpt.TrainInformation:JR-East.Yamanote"</code>	string (URL)
odpt:timeOfOrigin <i>required</i>	Occurrence time (ISO8601 date and time format) Example : <code>"2017-01-13T15:15:00+09:00"</code>	string (xsd:dateTime)
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:railway <i>optional</i>	ID representing the line where the operation information was generated (owl:sameAs of odpt:Railway) Example : <code>"odpt.Railway:JR-East.Yamanote"</code>	string (odpt:Railway owl:sameAs)
odpt:trainInformationStatus <i>optional</i>	Omitted during normal operation. If there is operation information, "運行情報あり" is stored. If information such as delay is available, store string such as "遅延" (multilingual-support) Example : <code>{"ja" : "日本語テキスト", "en" : "Text in English"}</code>	object
odpt:trainInformationText <i>required</i>	Operation information text (multilingual-support) Example : <code>{"ja" : "日本語テキスト", "en" : "Text in English"}</code>	object
odpt:railDirection <i>optional</i>	ID representing the direction to which the operation information is applicable. For the ID, use the value of owl:sameAs of odpt:RailDirection. Omit if not available. Example : <code>"odpt.RailDirection:Outbound"</code>	string (odpt:RailDirection owl:sameAs)
odpt:trainInformationArea <i>optional</i>	Area of occurrence. Omitted if not available (multilingual-support). Example : <code>{"ja" : "日本語テキスト", "en" : "Text in English"}</code>	object
odpt:trainInformationKind <i>optional</i>	Railway type. Omitted if not available (multilingual-support). Example : <code>{"ja" : "日本語テキスト", "en" : "Text in English"}</code>	object
odpt:stationFrom <i>optional</i>	Starting point of the location of occurrence. Omit if not available. Example : <code>"odpt.Station:JR-East.Yamanote.Tokyo"</code>	string (odpt:Station owl:sameAs)
odpt:stationTo <i>optional</i>	Ending point of the location of occurrence. Omitted if not available. Example : <code>"odpt.Station:JR-East.Yamanote.Tokyo"</code>	string (odpt:Station owl:sameAs)
odpt:trainInformationRange <i>optional</i>	Occurrence interval/range. Omitted if not available (multilingual-support). Example : <code>{"ja" : "日本語テキスト", "en" : "Text in English"}</code>	object
odpt:trainInformationCause <i>optional</i>	Reason for occurrence. Omit if not available (multilingual-support). Example : <code>{"ja" : "日本語テキスト", "en" : "Text in English"}</code>	object
odpt:transferRailways <i>optional</i>	A list of emergency transfer lines Example : <code>["odpt.Railway:JR-East.Yamanote"]</code>	< string (odpt:Railway owl:sameAs) > array
odpt:resumeEstimate <i>optional</i>	Estimated recovery time. However, this is often not disseminated. Example : <code>"2017-01-13T15:15:00+09:00"</code>	string (xsd:dateTime)

3.3.9. odpt:TrainTimetable

Shows the train timetable.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:TrainTimetable Example : <code>"odpt:TrainTimetable"</code>	string
dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dct:issued <i>optional</i>	Schedule revision date (ISO8601 date format) Example : <code>"2017-01-13"</code>	string (xsd:date)
dct:valid <i>optional</i>	Data guarantee period (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.TrainTimetable:operatorname.linename.trainnumber.thedayoftheweektype. If duplicates occur with the above naming rules, they are distinguished by adding a suffix, .1, .2, .3, etc. Example : <code>"odpt.TrainTimetable:JR-East.ChuoRapid.123M.Weekday"</code>	string (URL)
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:railway <i>required</i>	Stores the line ID. For the ID, use the value of owl:sameAs of odpt:Railway. Example : <code>"odpt.Railway:JR-East.Yamanote"</code>	string (odpt:Railway owl:sameAs)
odpt:railDirection <i>optional</i>	ID representing the direction of travel (owl:sameAs of odpt:RailDirection) Example : <code>"odpt.RailDirection:Outbound"</code>	string (odpt:RailDirection owl:sameAs)
odpt:calendar <i>optional</i>	ID of the day of the week / date information for the operation (owl:sameAs of odpt:Calendar) Example : <code>"odpt.Calendar:Weekday"</code>	string (odpt:Calendar owl:sameAs)
odpt:train <i>optional</i>	Train ID (owl:sameAs of odpt:Train) Example : <code>"odpt.Train:JR-East.Yamanote.123M"</code>	string (odpt:Train owl:sameAs)
odpt:trainNumber <i>required</i>	Train number Example : <code>"123M"</code>	string
odpt:trainType <i>optional</i>	Train type ID (owl:sameAs of odpt:TrainType) Example : <code>"odpt.TrainType:JR-East.Local"</code>	string (odpt:TrainType owl:sameAs)
odpt:trainName <i>optional</i>	A list of train service names and nicknames (multilingual-support) Example : <code>[{"ja" : "むさし", "en" : "Musashi"}]</code>	< object > array
odpt:trainOwner <i>optional</i>	ID representing the operator that owns the vehicle (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
odpt:originStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the starting station of the train service Example : <code>["odpt.Station:JR-East.ChuoRapid.Tokyo"]</code>	< string (odpt:Station owl:sameAs) > array

odpt:destinationStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the final station (destination station) of the train service Example : ["odpt.Station:JR-East.ChuoRapid.Takao"]	< string (odpt:Station owl:sameAs) > array
odpt:viaStation <i>optional</i>	A list of IDs (owl:sameAs of odpt:Station) representing the transfer-stations of the train service Example : ["odpt.Station:TokyoMetro.Tozai.NishiFunabashi"]	< string (odpt:Station owl:sameAs) > array
odpt:viaRailway <i>optional</i>	A list of IDs (owl:sameAs of odpt:Railway) representing the transmit lines where the train passes Example : ["odpt.Railway:TokyoMetro.Tozai"]	< string (odpt:Railway owl:sameAs) > array
odpt:previousTrainTimetable <i>optional</i>	When the train timetable of the same train is divided into multiple timetables, a list of ID (owl:sameAs of odpt:TrainTimetable) representing the immediately preceding train timetable Example : ["odpt.TrainTimetable:JR-East.ChuoRapid.123M.Weekday"]	< string (odpt:TrainTimetable owl:sameAs) > array
odpt:nextTrainTimetable <i>optional</i>	When the train timetable of the same train is divided into multiple timetables, a list of ID (owl:sameAs of odpt:TrainTimetable) representing the immediately succeeding train timetable Example : ["odpt.TrainTimetable:JR-East.ChuoRapid.123M.Weekday"]	< string (odpt:TrainTimetable owl:sameAs) > array
odpt:trainTimetableObject <i>required</i>	Stores a list of pairs of departure time and departure station or pairs of arrival time and arrival station Example : [{"odpt:departureTime" : "06:00", "odpt:departureStation" : "odpt.Station:JR-East.ChuoRapid.Tokyo"}, {"odpt:arrivalTime" : "07:00", "odpt:arrivalStation" : "odpt.Station:JR-East.ChuoRapid.Takao"}]	< odpt:trainTimetableObject > array
odpt:needExtraFee <i>optional</i>	Is an extra fee necessary in addition to the normal ticket? Example : true	boolean (xsd:bool)
odpt:note <i>optional</i>	Natural language description of annotation information not defined as properties (multilingual-support) Example : {"ja" : "日本語での注釈情報", "en" : "Note in English"}	object

odpt:trainTimetableObject

Name	Description	Schema
odpt:arrivalTime <i>optional</i>	Arrival time in ISO8601 time format. If the departure time and the arrival time are different, the arrival time may be described. Example : "05:08"	string (odpt:Time)
odpt:arrivalStation <i>optional</i>	ID of destination station (owl:sameAs of odpt:Station) Example : "odpt.Station:JR-East.ChuoRapid.Tokyo"	string (odpt:Station owl:sameAs)
odpt:departureTime <i>optional</i>	Departure time (ISO8601 time format) Example : "05:09"	string (odpt:Time)
odpt:departureStation <i>optional</i>	ID of the starting station (owl:sameAs of odpt:Station) Example : "odpt.Station:JR-East.ChuoRapid.Tokyo"	string (odpt:Station owl:sameAs)
odpt:platformNumber <i>optional</i>	Platform number where the train arrives Example : "1"	string
odpt:platformName <i>optional</i>	The name of the platform where the train arrives or departs (multilingual-support) Example : {"ja" : "日本語名称", "en" : "English Name"}	object
odpt:note	Natural language description of annotation information not defined as properties	

<i>optional</i>	(multilingual-support) Example : <code>{"ja" : "日本語での注釈情報", "en" : "Note in English"}</code>	object
-----------------	--	--------

3.3.10. odpt:TrainType

Defines the type of train, such as local or rapid

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:TrainType Example : <code>"odpt:TrainType"</code>	string
dc:date <i>optional</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.TrainType:operatorname.trainType. Example : <code>"odpt:JR-East.Local"</code>	string (URL)
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:JR-East"</code>	string (odpt:Operator owl:sameAs)
dc:title <i>optional</i>	Train type (Japanese) Example : <code>"普通"</code>	string
odpt:trainTypeTitle <i>optional</i>	Train type (multilingual-support) Example : <code>{"ja" : "普通", "en" : "Local"}</code>	object

4. ODPT Bus API

4.1. Overview

The Bus API can acquire the following six types of data related to buses.

- odpt:Bus
- odpt:BusstopPole
- odpt:BusroutePattern
- odpt:BusstopPoleTimetable
- odpt:BusroutePatternFare
- odpt:BusTimetable

For the contents of each data, refer to each explanation.

4.1.1. Relationship of each data

- The bus (odpt:Bus) is operated along a specific service route (odpt:BusroutePattern).
- The service route (odpt:BusroutePattern) is defined in the order of passing the stops (posts) (odpt:BusstopPole).
- The bus line (odpt:BusroutePattern odpt:busroute) may be composed of a plurality of routes (odpt:BusroutePattern).

- The routes on the same line have the same `odpt:BusroutePattern` `odpt:busroute`.
- The bus timetable (`odpt:BusTimetable`) is defined as a set of departure and arrival times at the stops (posts) in the order of visit of the stops of the service route (`odpt:BusroutePattern`).
- The bus stop (post) timetable (`odpt:BusstopPoleTimetable`) is a collection of bus timetables of buses that can be boarded from the bus stop (post) (`odpt:BusstopPole`).
 - A plurality of timetable data exists for one bus stop (post) for different days of the week (weekdays, Saturdays, etc.), different destinations and different routes.
- The fare is defined as fare information between a boarding post (`odpt:BusstopPole`) in a specific route and a departing post in a specific route.

4.1.2. Version information

Version : 4.0.0

4.1.3. URI scheme

Host : api.odpt.org

BasePath : /api/v4

Schemes : HTTPS

4.1.4. MIME type

- `application/json`

4.2. Paths

4.2.1. GET /api/v4/odpt:Bus

Description

Acquire the operation information (`odpt:Bus`) of the bus vehicle.

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:busroutePattern <i>optional</i>	ID of the route of the bus vehicle (owl:sameAs of <code>odpt:BusroutePattern</code>)	string (<code>odpt:BusroutePattern</code> owl:sameAs)
Query	odpt:operator <i>optional</i>	Transportation operator ID (owl:sameAs of <code>odpt:Operator</code>)	string (<code>odpt:Operator</code> owl:sameAs)
Query	odpt:fromBusstopPole <i>optional</i>	ID of the bus stop which the bus has just passed by or stopped at (owl:sameAs of <code>odpt:BusstopPole</code>)	string (<code>odpt:BusstopPole</code> owl:sameAs)
Query	odpt:toBusstopPole <i>optional</i>	ID of the bus stop which the bus visits next (owl:sameAs of <code>odpt:BusstopPole</code>)	string (<code>odpt:BusstopPole</code> owl:sameAs)

Responses

HTTP Code	Description	Schema

200	Successful completion	< odpt:Bus > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:Bus?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt_Bus.jsonld",
  "@type" : "odpt:Bus",
  "owl:sameAs" : "odpt.Bus:Toei.Ou57.40301.1.V366",
  "odpt:busroute" : "odpt.Busroute:Toei.Ou57",
  "odpt:operator" : "odpt.Operator:Toei",
  "odpt:busNumber" : "V366",
  "odpt:frequency" : 15,
  "odpt:busroutePattern" : "odpt.BusroutePattern:Toei.Ou57.40301.1",
  "odpt:fromBusstopPole" : "odpt.BusstopPole:Toei.Shimoicchoume.663.1",
  "odpt:fromBusstopPoleTime" : "2017-11-22T14:54:42+09:00",
  "odpt:toBusstopPole" : "odpt.BusstopPole:Toei.Kitashakoiriguchi.2294.1",
  "odpt:startingBusstopPole" : "odpt.BusstopPole:Toei.Akabaneekihigashic",
  "odpt:terminalBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoumedar",
  "dc:date" : "2017-11-22T14:56:49+09:00",
  "dct:valid" : "2017-11-22T14:57:04+09:00"
} ]
```

4.2.2. GET /api/v4/odpt:BusTimetable

Description

Acquire bus timetable (odpt:BusTimetable)

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:operator <i>required</i>	Transportation operator ID (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)

Query	odpt:busroutePattern <i>optional</i>	ID of the route on which the bus operates (owl:sameAs of odpt:BusroutePattern)	string (odpt:BusroutePattern owl:sameAs)
Query	dc:title <i>optional</i>	Bus line name (route name, etc.)	string
Query	odpt:calendar <i>optional</i>	Calendar ID (owl:sameAs of odpt:Calendar)	string (odpt:Calendar owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:BusTimetable > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:BusTimetable?acl:consumerKey=string&odpt:operator=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt_BusTimetable.jsonld",
  "@type" : "odpt:BusTimetable",
  "owl:sameAs" : "odpt:BusTimetable:SeibuBus.NeriDaka01.1001.1.1.Weekday",
  "dc:date" : "2017-11-15T09:57:34+09:00",
  "dc:title" : "練高01",
  "odpt:operator" : "odpt:Operator:SeibuBus",
  "odpt:busroutePattern" : "odpt:BusroutePattern:SeibuBus.NeriDaka01.1001.1.1.Weekday",
  "odpt:calendar" : "odpt:Calendar:Weekday",
  "odpt:busTimetableObject" : [ {
    "odpt:index" : 0,
    "odpt:busstopPole" : "odpt:BusstopPole:SeibuBus.Nerimatakanodaieki.2005",
    "odpt:departureTime" : "6:56",
    "odpt:destinationSign" : "成増駅南口",
    "odpt:isMidnight" : false,
    "odpt:canGetOn" : true,
    "odpt:canGetOff" : false
  }, {
    "odpt:index" : 1,
    "odpt:busstopPole" : "odpt:BusstopPole:SeibuBus.Takanodaicchoume.2005",
    "odpt:arrivalTime" : "6:57",
    "odpt:departureTime" : "6:57",
```

```
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 2,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Yaharasanchoume.20069.1",
"odpt:arrivalTime" : "6:59",
"odpt:departureTime" : "6:59",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 3,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokaminamiirigu",
"odpt:arrivalTime" : "7:00",
"odpt:departureTime" : "7:00",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 4,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokadainichuuga",
"odpt:arrivalTime" : "7:03",
"odpt:departureTime" : "7:03",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 5,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokaminamidoori",
"odpt:arrivalTime" : "7:03",
"odpt:departureTime" : "7:03",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 6,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Takamatsuyonchoume.200",
"odpt:arrivalTime" : "7:04",
"odpt:departureTime" : "7:04",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 7,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokasanchoume.2",
"odpt:arrivalTime" : "7:05",
"odpt:departureTime" : "7:05",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 8,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokaeki.20088.1",
"odpt:arrivalTime" : "7:09",
"odpt:departureTime" : "7:09",
```

```
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 9,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokaima.20089.1",
"odpt:arrivalTime" : "7:09",
"odpt:departureTime" : "7:09",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 10,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokadanchi.2009",
"odpt:arrivalTime" : "7:10",
"odpt:departureTime" : "7:10",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 11,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokarokuchoume.",
"odpt:arrivalTime" : "7:11",
"odpt:departureTime" : "7:11",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 12,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Asahichouminamichikuku",
"odpt:arrivalTime" : "7:12",
"odpt:departureTime" : "7:12",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 13,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokakoukou.2009",
"odpt:arrivalTime" : "7:13",
"odpt:departureTime" : "7:13",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 14,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Asahichounichoume.2009",
"odpt:arrivalTime" : "7:14",
"odpt:departureTime" : "7:14",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 15,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Hikarigaokakouenkita.2",
"odpt:arrivalTime" : "7:14",
"odpt:departureTime" : "7:14",
```

```
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 16,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Asahichoutoeijuutakuma
"odpt:arrivalTime" : "7:15",
"odpt:departureTime" : "7:15",
"odpt:destinationSign" : "成増駅南口",
"odpt:isMidnight" : false,
"odpt:canGetOn" : true,
"odpt:canGetOff" : true
}, {
"odpt:index" : 17,
"odpt:busstopPole" : "odpt.BusstopPole:SeibuBus.Narimasuekiminamiguchi
"odpt:arrivalTime" : "7:22",
"odpt:isMidnight" : "false,",
"odpt:canGetOn" : "false,",
"odpt:canGetOff" : true
} ]
} ]
```

4.2.3. GET /api/v4/odpt:BusroutePattern

Description

Acquisition of route information (odpt:BusroutePattern)

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	dc:title <i>optional</i>	Line / route name	string
Query	odpt:operator <i>optional</i>	Transportation operator ID (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:busroute <i>optional</i>	Line ID	string

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:BusroutePattern > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content

404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:BusroutePattern?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt_BusroutePattern.jsonld",
  "@type" : "odpt:BusroutePattern",
  "ow:sameAs" : "odpt:BusroutePattern:Toei.Ou57.40301.1",
  "dc:date" : "2017-11-14T17:45:28+09:00",
  "dc:title" : "王57",
  "odpt:kana" : "おう、ごじゅうなな",
  "odpt:operator" : "odpt:Operator:Toei",
  "odpt:busroute" : "odpt:Busroute:Toei.Ou57",
  "odpt:pattern" : "40301",
  "odpt:direction" : "1",
  "odpt:busstopPoleOrder" : [ {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Akabaneekihigashiguchi.21.1",
    "odpt:index" : 1
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Akabanenichoume.23.1",
    "odpt:index" : 2
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Iwabuchimachi.126.1",
    "odpt:index" : 3
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Shimonichoume.664.1",
    "odpt:index" : 4
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Shimoicchoume.663.1",
    "odpt:index" : 5
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Kitashakoiriguchi.2294.1",
    "odpt:index" : 6
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Kitashakomae.407.2",
    "odpt:index" : 7
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Kamiyarikkyou.2290.3",
    "odpt:index" : 8
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Kitakukamiyachou.403.1",
    "odpt:index" : 9
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Oujigochoume.201.1",
    "odpt:index" : 10
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Oujiyonchoume.200.1",
    "odpt:index" : 11
  }, {
    "odpt:busstopPole" : "odpt:BusstopPole:Toei.Oujisanchoume.199.1",
    "odpt:index" : 12
  }
]
```

```

        "odpt:index" : 12
      }, {
        "odpt:busstopPole" : "odpt.BusstopPole:Toei.Oujinichoume.198.1",
        "odpt:index" : 13
      }, {
        "odpt:busstopPole" : "odpt.BusstopPole:Toei.Oujiekimae.196.10",
        "odpt:index" : 14
      }, {
        "odpt:busstopPole" : "odpt.BusstopPole:Toei.Toshimanichoume.1001.2",
        "odpt:index" : 15
      }, {
        "odpt:busstopPole" : "odpt.BusstopPole:Toei.Toshimasanchoume.1002.2",
        "odpt:index" : 16
      }, {
        "odpt:busstopPole" : "odpt.BusstopPole:Toei.Toshimayonchoume.1003.1",
        "odpt:index" : 17
      }, {
        "odpt:busstopPole" : "odpt.BusstopPole:Toei.Toshimarokuchoume.1007.2",
        "odpt:index" : 18
      }, {
        "odpt:busstopPole" : "odpt.BusstopPole:Toei.Toshimagochoumedanchi.1004",
        "odpt:index" : 19
      }
    ]
  }
]

```

4.2.4. GET /api/v4/odpt:BusroutePatternFare

Description

Acquire fare information (odpt:BusroutePatternFare)

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:operator <i>optional</i>	Transportation operator ID (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:fromBusstopPole <i>optional</i>	ID of the boarding bus stop (post) (owl:sameAs of odpt:BusstopPole)	string (odpt:BusstopPole owl:sameAs)
Query	odpt:toBusstopPole <i>optional</i>	ID of the departing bus stop (post) owl:sameAs of odpt:BusstopPole	string (odpt:BusstopPole owl:sameAs)
Query	odpt:ticketFare <i>optional</i>	Fare for ticket	integer
Query	odpt:childTicketFare <i>optional</i>	Child fare for ticket	integer
Query	odpt:icCardFare <i>optional</i>	Fare for IC card	integer
Query	odpt:childIcCardFare	Child fare for IC card	integer

	<i>optional</i>	
--	-----------------	--

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:BusroutePatternFare > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:BusroutePatternFare?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt_BusroutePatternFare.json",
  "@type" : "odpt:BusroutePatternFare",
  "owl:sameAs" : "odpt:BusroutePatternFare:Toei.Ou57.40301.1.1.Akabaneeki",
  "dc:date" : "2017-11-14T17:45:31+09:00",
  "odpt:operator" : "odpt:Operator:Toei",
  "odpt:fromBusroutePattern" : "odpt:BusroutePattern:Toei.Ou57.40301.1",
  "odpt:fromBusstopPoleOrder" : 1,
  "odpt:fromBusstopPole" : "odpt:BusstopPole:Toei.Akabaneekihigashiguchi",
  "odpt:toBusroutePattern" : "odpt:BusroutePattern:Toei.Ou57.40301.1",
  "odpt:toBusstopPoleOrder" : 2,
  "odpt:toBusstopPole" : "odpt:BusstopPole:Toei.Akabanenichoume.23.1",
  "odpt:ticketFare" : 210,
  "odpt:childTicketFare" : 110,
  "odpt:icCardFare" : 206,
  "odpt:childIcCardFare" : 103
} ]
```

4.2.5. GET /api/v4/odpt:BusstopPole

Description

Acquire bus stop (post) (odpt:BusstopPole)

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
	@id	Unique identifier assigned to data	

Query	<i>optional</i>		string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	dc:title <i>optional</i>	Bus stop name	string
Query	odpt:busstopPoleNumber <i>optional</i>	Bus stop post number	string
Query	odpt:platformNumber <i>optional</i>	Bus stop platform number	string
Query	odpt:busroutePattern <i>optional</i>	List of IDs of the route that departs and arrives at the post (owl:sameAs of odpt:BusroutePattern)	string (odpt:BusroutePattern owl:sameAs)
Query	odpt:operator <i>optional</i>	Transportation operator ID (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:BusstopPole > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:BusstopPole?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt_BusstopPole.jsonld",
  "@type" : "odpt:BusstopPole",
  "owl:sameAs" : "odpt.BusstopPole:Toei.Akabaneekihigashiguchi.21.1",
  "dc:date" : "2017-11-14T17:44:05+09:00",
  "dc:title" : "赤羽駅東口",
  "odpt:kana" : "あかばねえぎひがしぐち",
  "geo:long" : 139.7214941,
  "geo:lat" : 35.7790549,
  "odpt:busroutePattern" : [ "odpt.BusroutePattern:Toei.Ou57.40301.1",
  "odpt:operator" : [ "odpt.Operator:Toei" ],
```



```
"odpt:busstopPoleNumber" : "1",
"odpt:busstopTimetable" : [ "odpt.BusstopPoleTimetable:Toei.Ou57.Akaba"
} ]
```

4.2.6. GET /api/v4/odpt:BusstopPoleTimetable

Description

Acquire bus stop (post) timetable (odpt:BusstopPoleTimetable)

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:busstopPole <i>optional</i>	ID of the bus stop (post) of the timetable (owl:sameAs of odpt:BusstopPole)	string (odpt:BusstopPole owl:sameAs)
Query	odpt:busDirection <i>optional</i>	ID of the area	string
Query	odpt:busroute <i>optional</i>	Line ID	string
Query	odpt:operator <i>optional</i>	Transportation operator ID (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:calendar <i>optional</i>	Calendar ID (owl:sameAs of odpt:Calendar)	string (odpt:Calendar owl:sameAs)
Query	dc:date <i>optional</i>	Data generation date	string (xsd:dateTime)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:BusstopPoleTimetable > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:BusstopPoleTimetable?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt_BusstopPoleTimetable.",
  "@type" : "odpt:BusstopPoleTimetable",
  "owl:sameAs" : "odpt:BusstopPoleTimetable:Toei.Ou57.Akabaneekihigashiguchi.21.",
  "dc:date" : "2017-11-14T17:29:41+09:00",
  "dc:title" : "王 5 7:赤羽駅東口:豊島五丁目団地行:平日",
  "odpt:busstopPole" : "odpt:BusstopPole:Toei.Akabaneekihigashiguchi.21.",
  "odpt:busDirection" : "odpt:BusDirection:Toei.Toshimagochoumedanchi",
  "odpt:busroute" : "odpt:Busroute:Toei.Ou57",
  "odpt:operator" : "odpt:Operator:Toei",
  "odpt:calendar" : "odpt:Calendar:Weekday",
  "odpt:busstopPoleTimetableObject" : [ {
    "odpt:departureTime" : "06:21",
    "odpt:destinationBusstopPole" : "odpt:BusstopPole:Toei.Toshimagochoume",
    "odpt:destinationSign" : "豊島五丁目団地行",
    "odpt:isNonStepBus" : true,
    "odpt:isMidnight" : false
  }, {
    "odpt:departureTime" : "06:35",
    "odpt:destinationBusstopPole" : "odpt:BusstopPole:Toei.Toshimagochoume",
    "odpt:destinationSign" : "豊島五丁目団地行",
    "odpt:isNonStepBus" : true,
    "odpt:isMidnight" : false
  }, {
    "odpt:departureTime" : "06:52",
    "odpt:destinationBusstopPole" : "odpt:BusstopPole:Toei.Toshimagochoume",
    "odpt:destinationSign" : "豊島五丁目団地行",
    "odpt:isNonStepBus" : true,
    "odpt:isMidnight" : false
  }, {
    "odpt:departureTime" : "07:05",
    "odpt:destinationBusstopPole" : "odpt:BusstopPole:Toei.Toshimagochoume",
    "odpt:destinationSign" : "豊島五丁目団地行",
    "odpt:isNonStepBus" : true,
    "odpt:isMidnight" : false
  }, {
    "odpt:departureTime" : "07:17",
    "odpt:destinationBusstopPole" : "odpt:BusstopPole:Toei.Toshimagochoume",
    "odpt:destinationSign" : "豊島五丁目団地行",
    "odpt:isNonStepBus" : true,
    "odpt:isMidnight" : false
  }, {
    "odpt:departureTime" : "07:29",
    "odpt:destinationBusstopPole" : "odpt:BusstopPole:Toei.Toshimagochoume",
    "odpt:destinationSign" : "豊島五丁目団地行",
    "odpt:isNonStepBus" : true,
    "odpt:isMidnight" : false
  }, {
    "odpt:departureTime" : "07:39",
    "odpt:destinationBusstopPole" : "odpt:BusstopPole:Toei.Toshimagochoume",
    "odpt:destinationSign" : "豊島五丁目団地行",
    "odpt:isNonStepBus" : true,
    "odpt:isMidnight" : false
  }, {
    "odpt:departureTime" : "07:49",
    "odpt:destinationBusstopPole" : "odpt:BusstopPole:Toei.Toshimagochoume"
```

```
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "08:00",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "08:10",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "08:21",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "08:32",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "08:44",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "08:55",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "09:07",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "09:18",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "09:30",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "09:42",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
```

```
    }, {
      "odpt:departureTime" : "09:54",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "10:07",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "10:19",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "10:28",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "10:40",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "10:52",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "11:08",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "11:24",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "11:38",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "11:52",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "12:06",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume"
```

```
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "12:19",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "12:35",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "12:50",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "12:50",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "13:04",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "13:18",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "13:32",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "13:46",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "14:01",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "14:16",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
```

```
    }, {
      "odpt:departureTime" : "14:32",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "14:47",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "15:01",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "15:15",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "15:30",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "15:46",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "16:02",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "16:17",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "16:31",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "16:45",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "17:00",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume"
```

```
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "17:15",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "17:30",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "17:45",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "18:00",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "18:15",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "18:30",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "18:45",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "19:00",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "19:14",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
}, {
"odpt:departureTime" : "19:34",
"odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume
"odpt:destinationSign" : "豊島五丁目団地行",
"odpt:isNonStepBus" : true,
"odpt:isMidnight" : false
```

```
    }, {
      "odpt:departureTime" : "19:54",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "20:15",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "20:36",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "21:00",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "21:29",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "21:52",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "22:12",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }, {
      "odpt:departureTime" : "22:40",
      "odpt:destinationBusstopPole" : "odpt.BusstopPole:Toei.Toshimagochoume",
      "odpt:destinationSign" : "豊島五丁目団地行",
      "odpt:isNonStepBus" : true,
      "odpt:isMidnight" : false
    }
  ]
}
```

4.3. Definitions

4.3.1. odpt:Bus

odpt:Bus is a className that represents bus operation status information.

odpt:busroutePattern represents the route, and odpt:fromBusstopPole and odpt:toBusstopPole represent the current location.

Current location	odpt:fromBusstopPole	odpt:toBusstopPole
Between bus stop A and bus stop B	Bus stop A	Bus stop B

Stopping at bus stop A	Bus stop A	null
Approaching bus stop B	null	Bus stop B

If the information is not available regarding whether the bus approaches a bus stop, `odpt:fromBusstopPole` may not be null.

If the information is not available regarding whether the bus is stopped, `odpt:toBusstopPole` may not be null.

`odpt:occupancyStatus` represents the occupancy status of the vehicle.

Domain	Description
<code>odpt.OccupancyStatus:Empty</code>	Vehicle is empty.
<code>odpt.OccupancyStatus:ManySeatsAvailable</code>	Vehicle has many vacant seats.
<code>odpt.OccupancyStatus:FewSeatsAvailable</code>	Vehicle has some vacant seats.
<code>odpt.OccupancyStatus:StandingRoomOnly</code>	Vehicle has no vacant seats, but many standing space.
<code>odpt.OccupancyStatus:CrushedStandingRoomOnly</code>	Vehicle has no vacant seats, but some standing space.
<code>odpt.OccupancyStatus:FullRoomOnly</code>	Vehicle is almost full.
<code>odpt.OccupancyStatus:NotAcceptingPassengers</code>	Vehicle accepts no passengers.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier Example : <code>"string"</code>	string (URN)
@type <i>required</i>	The bus operation status information className name "odpt:Bus" is returned. Example : <code>"odpt:Bus"</code>	string
owl:sameAs <i>required</i>	Unique identifier of bus operation status information Example : <code>"string"</code>	string (URL)
odpt:busNumber <i>required</i>	Bus vehicle number Example : <code>"string"</code>	string
dc:date <i>required</i>	Data generation time Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dct:valid <i>required</i>	Data guarantee period Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
odpt:frequency <i>required</i>	The update frequency (seconds), the latest operation status is acquired by making a request after the specified number of seconds elapsed. Example : <code>60</code>	integer
odpt:busroutePattern <i>required</i>	ID of the route (owl:sameAs of <code>odpt:BusroutePattern</code>) Example : <code>"string"</code>	string (odpt:BusroutePattern owl:sameAs)
odpt:busTimetable <i>optional</i>	ID of the timetable (owl:sameAs of <code>odpt:BusTimetable</code>) Example : <code>"string"</code>	string (odpt:BusTimetable owl:sameAs)
	ID of the transportation operator (owl:sameAs of	

odpt:operator <i>required</i>	odpt:Operator) Example : <code>"string"</code>	string (odpt:Operator owl:sameAs)
odpt:startingBusstopPole <i>optional</i>	ID representing the first bus stop of the route (owl:sameAs of odpt:BusstopPole) Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:terminalBusstopPole <i>optional</i>	ID representing the last bus stop of the route (owl:sameAs of odpt:BusstopPole) Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:fromBusstopPole <i>optional</i>	ID of the bus stop which the bus has just passed by or stopped at (owl:sameAs of odpt:BusstopPole). See the beginning of the description of odpt:Bus for details. Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:fromBusstopPoleTime <i>optional</i>	The time of departure from the bus stop which the bus visited most recently. If odpt:fromBusstopPole is null, odpt:fromBusstopPoleTime is also null Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
odpt:toBusstopPole <i>optional</i>	ID of the bus stop which the bus visits next (owl:sameAs in odpt:BusstopPole). See the beginning of the description of odpt:Bus for details. Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:progress <i>optional</i>	Current position (ratio) where "From" is 0 and "To" is 1 Example : <code>0.0</code>	number (double)
geo:long <i>optional</i>	Longitude of the bus (decimal notation, geodetic system is WGS84) Example : <code>139.1234</code>	number (double)
geo:lat <i>optional</i>	Latitude of the bus (decimal notation, geodetic system is WGS84) Example : <code>35.1234</code>	number (double)
odpt:speed <i>optional</i>	Bus speed (km / hour) Example : <code>0.0</code>	number (double)
odpt:azimuth <i>optional</i>	Represents the azimuth of the traveling direction of the bus. The unit is degree (°). North is 0 degrees and it increases clockwise (eastward). Example : <code>90.0</code>	number (double)
odpt:doorStatus <i>optional</i>	Shows the open / closed status of the door of the bus. "open" is open, "close" is closed, "self" is for semi-automatic doors. Example : <code>"string"</code>	enum (open, close, self)
odpt:occupancyStatus <i>optional</i>	Occupancy status of the vehicle Example : <code>"odpt.OccupancyStatus:Empty"</code>	enum (odpt.OccupancyStatus:Empty, odpt.OccupancyStatus:ManySeatsAvailable, odpt.OccupancyStatus:FewSeatsAvailable, odpt.OccupancyStatus:StandingRoomOnly, odpt.OccupancyStatus:CrushedStandingRoomOnly, odpt.OccupancyStatus:FullRoomOnly, odpt.OccupancyStatus:NotAcceptingPassengers)

4.3.2. odpt:BusTimetable

It is a bus timetable.

Name	Description	Schema

@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	The className name of the bus timetable, "odpt:BusTimetable" is returned. Example : <code>"odpt:BusTimetable"</code>	string
owl:sameAs <i>required</i>	Unique identifier of the bus timetable Example : <code>"string"</code>	string (URL)
dc:date <i>optional</i>	Data generation time Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dct:issued <i>optional</i>	Schedule revision date Example : <code>"2017-01-13"</code>	string (xsd:date)
dct:valid <i>optional</i>	Data guarantee period Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dc:title <i>optional</i>	Bus line name (route name, etc.) Example : <code>"string"</code>	string
odpt:kana <i>optional</i>	Reading of bus line name (yomigana) Example : <code>"string"</code>	string
odpt:operator <i>required</i>	ID of the transportation operator (owl:sameAs of odpt:Operator) Example : <code>"string"</code>	string (odpt:Operator owl:sameAs)
odpt:busroutePattern <i>required</i>	ID of the route on which the bus operates (owl:sameAs of odpt:BusroutePattern) Example : <code>"string"</code>	string (odpt:BusroutePattern owl:sameAs)
odpt:calendar <i>required</i>	Calendar ID (owl:sameAs of odpt:Calendar) Example : <code>"string"</code>	string (odpt:Calendar owl:sameAs)
odpt:busTimetableObject <i>required</i>	Bus timetable hour and minute information Example : <code>["object"]</code>	< odpt:busTimetableObject > array

odpt:busTimetableObject

Name	Description	Schema
odpt:index <i>required</i>	Post passing order Example : <code>1</code>	integer
odpt:busstopPole <i>required</i>	Bus stop (post) ID (owl:sameAs of odpt:BusstopPole) Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:arrivalTime <i>optional</i>	Bus arrival time Example : <code>"10:00"</code>	string (odpt:Time)
odpt:departureTime <i>optional</i>	Bus departure time Example : <code>"10:00"</code>	string (odpt:Time)
odpt:destinationSign <i>optional</i>	Destination (front display) information Example : <code>"string"</code>	string

odpt:isNonStepBus <i>optional</i>	True for non-step bus Example : <code>true</code>	boolean
odpt:isMidnight <i>optional</i>	True for late-night bus Example : <code>true</code>	boolean
odpt:canGetOn <i>optional</i>	True if rideable Example : <code>true</code>	boolean
odpt:canGetOff <i>optional</i>	True if you can get off Example : <code>true</code>	boolean
odpt:note <i>optional</i>	Note Example : <code>"string"</code>	string

4.3.3. odpt:BusroutePattern

Bus route information `odpt:BusroutePattern` represents the route information of the bus line. `odpt:busstopPoleOrder` expresses the order of stops (posts) where buses stop. The bus line ('㊦ 57', etc.) is usually composed of a plurality of routes. (E.g. outbound, inbound, different stop order variation)

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	"odpt:BusroutePattern", the className name of the bus line information is returned. Example : <code>"odpt:BusroutePattern"</code>	string
owl:sameAs <i>required</i>	Unique identifier of bus line information Example : <code>"string"</code>	string (URL)
dc:date <i>required</i>	Data generation time Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dct:valid <i>optional</i>	Data guarantee period Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dc:title <i>required</i>	Bus line name (route name, route number, etc.) Example : <code>"string"</code>	string
odpt:kana <i>optional</i>	Reading of bus line name (yomigana) Example : <code>"string"</code>	string
odpt:operator <i>required</i>	ID (owl:sameAs of odpt:Operator) representing transportation operator Example : <code>"string"</code>	string (odpt:Operator owl:sameAs)
odpt:busroute <i>optional</i>	ID representing the route Example : <code>"string"</code>	string
odpt:pattern <i>optional</i>	Route pattern Example : <code>"string"</code>	string
odpt:direction <i>optional</i>	direction Example : <code>"string"</code>	string

ug:region <i>optional</i>	Feature information in GeoJSON format Example : <code>"object"</code>	object
odpt:busstopPoleOrder <i>required</i>	Order of bus stops (posts) Example : <code>["object"]</code>	< odpt:busstopPoleOrder > array
odpt:note <i>optional</i>	Note Example : <code>"string"</code>	string
odpt:busLocationURL <i>optional</i>	Website URL that provides bus location information Example : <code>"string"</code>	string (URL)

odpt:busstopPoleOrder

Name	Description	Schema
odpt:busstopPole <i>required</i>	Bus stop ID (owl:sameAs of odpt:BusstopPole) Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:index <i>required</i>	Order of bus stops. It is in the order of visits. Example : <code>1</code>	integer
odpt:openingDoorsToGetOn <i>optional</i>	Doors used for boarding Example : <code>["string"]</code>	< enum (odpt:OpeningDoor:FrontSide, odpt:OpeningDoor:RearSide) > array
odpt:openingDoorsToGetOff <i>optional</i>	Doors used for getting off Example : <code>["string"]</code>	< enum (odpt:OpeningDoor:FrontSide, odpt:OpeningDoor:RearSide) > array
odpt:note <i>optional</i>	Note Example : <code>"string"</code>	string

4.3.4. odpt:BusroutePatternFare

Fare information between boarding bus stop (post) and departing bus stop (post)

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	The bus fare className name "odpt:BusroutePatternFare" is returned here. Example : <code>"odpt:BusroutePatternFare"</code>	string
owl:sameAs <i>required</i>	Unique identifier of bus fare Example : <code>"string"</code>	string (URL)
dc:date <i>required</i>	Data generation time Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dct:issued <i>optional</i>	Fare revision date Example : <code>"2017-01-13"</code>	string (xsd:date)
dct:valid <i>optional</i>	Data guarantee period Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)

odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"string"</code>	string (odpt:Operator owl:sameAs)
odpt:fromBusroutePattern <i>required</i>	ID representing the route pattern (owl:sameAs of odpt:BusroutePattern) Example : <code>"string"</code>	string (odpt:BusroutePattern owl:sameAs)
odpt:fromBusstopPoleOrder <i>required</i>	The order of bus stops (posts) visited in the route pattern accessed from the boarding stop. Same value as odpt:index of odpt:busstopPoleOrder of odpt:BusroutePattern represented in odpt:fromBusroutePattern. Example : <code>0</code>	integer
odpt:fromBusstopPole <i>required</i>	ID representing the boarding bus stop. Same as the ID of the bus stop (post) represented by odpt:fromBusroutePattern, odpt:fromBusstopPoleOrder. Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:toBusroutePattern <i>required</i>	ID representing the departing route pattern (owl:sameAs of odpt:BusroutePattern) Example : <code>"string"</code>	string (odpt:BusroutePattern owl:sameAs)
odpt:toBusstopPoleOrder <i>required</i>	The order of bus stops (posts) visited in the route pattern accessed from the departing stop. Same value as odpt:index of odpt:busstopPoleOrder of odpt:BusroutePattern represented by odpt:toBusroutePattern. Example : <code>0</code>	integer
odpt:toBusstopPole <i>required</i>	ID representing the departing bus stop (owl:sameAs of odpt:BusstopPole). Same as the ID of the bus stop (post) represented in odpt:toBusroutePattern, odpt:toBusstopPoleOrder. Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:ticketFare <i>required</i>	Fare for ticket (yen) Example : <code>200</code>	integer
odpt:childTicketFare <i>optional</i>	Child fare for ticket (yen) Example : <code>100</code>	integer
odpt:icCardFare <i>optional</i>	Fare for IC card (yen) Example : <code>200</code>	integer
odpt:childIcCardFare <i>optional</i>	Child fare for IC card (yen) Example : <code>100</code>	integer

4.3.5. odpt:BusstopPole

Bus stop information:odpt:BusstopPole represents bus stop (post) information.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	The className name of the bus stop (post), "odpt:BusstopPole" is returned here. Example : <code>"odpt:BusstopPole"</code>	string
owl:sameAs <i>required</i>	Unique identifier of bus stop (post) Example : <code>"string"</code>	string (URL)

dc:date <i>required</i>	Data generation time Example : "2017-01-13T15:10:00+09:00"	string (xsd:dateTime)
dct:valid <i>optional</i>	Data guarantee period Example : "2017-01-13T15:10:00+09:00"	string (xsd:dateTime)
dc:title <i>required</i>	Bus stop name Example : "豊洲駅前"	string
odpt:kana <i>optional</i>	Reading of bus stop name (yomigana) Example : "とよすえきまえ"	string
title <i>optional</i>	Bus stop name (multilingual-support) Example : {"ja" : "豊洲駅前", "ja-Hrkt" : "とよすえきまえ", "en" : "Toyosu Sta."}	object
geo:long <i>optional</i>	Sign post longitude (WGS84) Example : 139.1234	number (double)
geo:lat <i>optional</i>	Sign post latitude (WGS84) Example : 35.1234	number (double)
ug:region <i>optional</i>	Feature information in GeoJSON format Example : "object"	object
odpt:busroutePattern <i>optional</i>	A list of IDs of route patterns that come to this bus stop. (owl:sameAs of odpt:BusroutePattern) Example : ["string"]	< string (odpt:BusroutePattern owl:sameAs) > array
odpt:operator <i>required</i>	A list of IDs (owl:sameAs of odpt:Operator) representing the operators of the incoming buses Example : ["string"]	< string (odpt:Operator owl:sameAs) > array
odpt:busstopPoleNumber <i>optional</i>	Sign post number. It distinguishes different posts at the same stop, and there is no guarantee that sign post number matches the bus stop number. Example : "string"	string
odpt:platformNumber <i>optional</i>	Bus stop platform number Example : "string"	string
odpt:busstopPoleTimetable <i>optional</i>	A list of IDs (owl:sameAs of odpt:BusstopPoleTimetable) of bus stop (post) timetables Example : ["string"]	< string (odpt:BusstopPoleTimetable owl:sameAs) > array

4.3.6. odpt:BusstopPoleTimetable

The bus stop (pole) timetable represented by odpt:busstopPole. The data may be different depending on the day of the week, destination, line, and the like.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : "http://vocab.odpt.org/context_odpt.jsonld"	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : "urn:ucode:_00001C0000000000000010000030FD7E5"	string (URN)
	"odpt:BusstopPoleTimetable", in the bus stop (post)	

@type <i>required</i>	timetable className name is returned here. Example : <code>"odpt:BusstopPoleTimetable"</code>	string
owl:sameAs <i>required</i>	Unique identifier of bus stop (post) timetable Example : <code>"string"</code>	string (URL)
dc:date <i>required</i>	Data generation time Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dct:issued <i>optional</i>	Schedule revision date Example : <code>"2017-01-13"</code>	string (xsd:date)
dct:valid <i>optional</i>	Data guarantee expiration date. ISO8601 format. Stored only when an expiration date exists. Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
dc:title <i>optional</i>	Bus line name (route name, etc.) Example : <code>"string"</code>	string
odpt:busstopPole <i>required</i>	ID representing the bus stop (post) (owl:sameAs of odpt:BusstopPole) Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:busDirection <i>required</i>	ID representing the directions. It may be an array. Example : <code>"string"</code>	string
odpt:busroute <i>required</i>	ID representing the line. It may be an array. (In case of a timetable listing multiple lines) Example : <code>"string"</code>	string
odpt:operator <i>required</i>	ID representing the operator (owl:sameAs of odpt:Operator) Example : <code>"string"</code>	string (odpt:Operator owl:sameAs)
odpt:calendar <i>required</i>	Day of the week and the date of operation (owl:sameAs of odpt:Calendar) Example : <code>"string"</code>	string (odpt:Calendar owl:sameAs)
odpt:busstopPoleTimetableObject <i>optional</i>	Hour and minute information of bus stop (post) timetable Example : <code>["object"]</code>	< odpt:busstopPoleTimetableObject > array

odpt:busstopPoleTimetableObject

Name	Description	Schema
odpt:arrivalTime <i>optional</i>	Bus arrival time Example : <code>"10:00"</code>	string (odpt:Time)
odpt:departureTime <i>required</i>	Bus departure time Example : <code>"10:00"</code>	string (odpt:Time)
odpt:destinationBusstopPole <i>optional</i>	ID of the destination bus stop (post) (owl:sameAs of odpt:BusstopPole) Example : <code>"string"</code>	string (odpt:BusstopPole owl:sameAs)
odpt:destinationSign <i>optional</i>	Destination (front display) information Example : <code>"string"</code>	string
odpt:busroutePattern <i>optional</i>	ID of the bus line (owl:sameAs of odpt:BusroutePattern) Example : <code>"string"</code>	string (odpt:BusroutePattern owl:sameAs)

odpt:busroutePatternOrder <i>optional</i>	The order of bus stops (posts) visited in the route pattern. Same value as odpt:index of odpt:busstopPoleOrder of odpt:BusroutePattern represented in odpt:BusroutePattern. Example : <code>0</code>	integer
odpt:isNonStepBus <i>optional</i>	True for non-step bus Example : <code>true</code>	boolean
odpt:isMidnight <i>optional</i>	True for late-night bus Example : <code>true</code>	boolean
odpt:canGetOn <i>optional</i>	True if rideable Example : <code>true</code>	boolean
odpt:canGetOff <i>optional</i>	True if you can get off Example : <code>true</code>	boolean
odpt:note <i>optional</i>	Note Example : <code>"string"</code>	string

5. ODPT Airplane API

5.1. Overview

With the Airplane API, the following data related to air flights can be acquired.

- odpt:Airport
- odpt:AirportTerminal
- odpt:FlightInformationArrival
- odpt:FlightInformationDeparture
- odpt:FlightSchedule
- odpt:FlightStatus

For the contents of each data, refer to each explanation.

5.1.1. Version information

Version : 4.0.0

5.1.2. URI scheme

Host : api.odpt.org

BasePath : /api/v4

Schemes : HTTPS

5.1.3. MIME type

- `application/json`

5.2. Paths

5.2.1. GET /api/v4/odpt:Airport

Description

Acquire airport information.

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:Airport > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:Airport?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:ucode:_00001C000000000000000100000030FD7E5",
  "@type" : "odpt:Airport",
  "dc:date" : "2017-01-13T15:10:00+09:00",
  "owl:sameAs" : "odpt.Airport:HND",
  "dc:title" : "東京(羽田)",
  "odpt:airportTitle" : {
    "ja" : "東京(羽田)",
    "en" : "Tokyo (Haneda)"
  },
  "odpt:airportTerminal" : [ "odpt.AirportTerminal:HND.DomesticTerminal1"
] }
```

5.2.2. GET /api/v4/odpt:AirportTerminal

Description

Acquire airport terminal information.

Parameters

Type	Name	Description	Schema
------	------	-------------	--------

Responses

Example HTTP request

```
/api/v4/odpt:AirportTerminal?acl:consumerKey=string
```

Response 200

```
[ {
    "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
    "@id" : "urn:ucode:_00001C000000000000010000030FD7E5",
    "@type" : "odpt:AirportTerminal",
    "dc:date" : "2017-01-13T15:10:00+09:00",
    "owl:sameAs" : "odpt.AirportTerminal:HND.DomesticTerminal1",
    "dc:title" : "国内線第1ターミナル",
    "odpt:airportTerminalTitle" : {
        "ja" : "国内線第1ターミナル",
        "en" : "Domestic Terminal 1"
    },
    "odpt:airport" : "odpt.Airport:HND"
} ]
```

Description

Acquire flight arrival information (real-time information of air flights arriving at the airport on the given day).

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:operator <i>optional</i>	ID representing the airport operator or airline operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:airline <i>optional</i>	ID representing the airline operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:flightStatus <i>optional</i>	ID representing the flight status (owl:sameAs of odpt:FlightStatus)	string (odpt:FlightStatus owl:sameAs)
Query	odpt:arrivalAirport <i>optional</i>	ID representing the arrival airport (owl:sameAs of odpt:Airport)	string (odpt:Airport owl:sameAs)
Query	odpt:arrivalAirportTerminal <i>optional</i>	ID representing the terminal at the arrival airport (owl:sameAs of odpt:AirportTerminal)	string (odpt:AirportTerminal owl:sameAs)
Query	odpt:arrivalGate <i>optional</i>	Arrival airport gate number	string
Query	odpt:originAirport <i>optional</i>	ID representing the departure airport (owl:sameAs of odpt:Airport)	string (odpt:Airport owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:FlightInformationArrival > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:FlightInformationArrival?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
```

```
"@context" : "http://vocab.odpt.org/context_odpt.jsonld",
"@id" : "urn:uuid:ff91feee-2d2c-4365-902e-26375fdf9d8b",
"@type" : "odpt:FlightInformationArrival",
"dc:date" : "2017-12-06T23:15:01+09:00",
"owl:sameAs" : "odpt:FlightInformationArrival:NAA.NRT.NH832",
"odpt:operator" : "odpt.Operator:NAA",
"odpt:airline" : "odpt.Operator:ANA",
"odpt:flightNumber" : [ "NH832" ],
"odpt:flightStatus" : "odpt.FlightStatus:Arrived",
"odpt:scheduledArrivalTime" : "06:45",
"odpt:actualArrivalTime" : "06:48",
"odpt:arrivalAirport" : "odpt.Airport:NRT",
"odpt:arrivalAirportTerminal" : "odpt.AirportTerminal:NRT.Terminal1",
"odpt:arrivalGate" : "27",
"odpt:baggageClaim" : "27",
"odpt:originAirport" : "odpt.Airport:SGN",
"odpt:aircraftType" : "788"

} ]
```

5.2.4. GET /api/v4/odpt:FlightInformationDeparture

Description

Acquire flight departure information (real-time information of the flights departing the airport on the given day).

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
Query	odpt:operator <i>optional</i>	ID representing the airport operator or airline operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:airline <i>optional</i>	ID representing the airline operator (owl:sameAs of odpt:Operator)	string (odpt:Operator owl:sameAs)
Query	odpt:flightStatus <i>optional</i>	ID representing the flight status (owl:sameAs of odpt:FlightStatus)	string (odpt:FlightStatus owl:sameAs)
Query	odpt:departureAirport <i>optional</i>	ID representing the departure airport (owl:sameAs of odpt:Airport)	string (odpt:Airport owl:sameAs)
Query	odpt:departureAirportTerminal <i>optional</i>	ID representing the terminal at the departure airport (owl:sameAs of odpt:AirportTerminal)	string (odpt:AirportTerminal owl:sameAs)
Query	odpt:departureGate <i>optional</i>	Departure airport gate number	string
Query	odpt:destinationAirport <i>optional</i>	ID representing the destination airport (owl:sameAs of odpt:Airport)	string (odpt:Airport owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:FlightInformationDeparture > array

400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:FlightInformationDeparture?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:uuid:f476346b-ae6d-4102-99d8-2bf80d7c4dd8",
  "@type" : "odpt:FlightInformationDeparture",
  "dc:date" : "2017-12-06T23:20:02+09:00",
  "owl:sameAs" : "odpt.FlightInformationDeparture:NAA.NRT.9W4807",
  "odpt:operator" : "odpt.Operator:NAA",
  "odpt:airline" : "odpt.Operator:JAI",
  "odpt:flightNumber" : [ "9W4807" ],
  "odpt:flightStatus" : "odpt.FlightStatus:Takeoff",
  "odpt:scheduledDepartureTime" : "08:30",
  "odpt:actualDepartureTime" : "08:32",
  "odpt:departureAirport" : "odpt.Airport:NRT",
  "odpt:departureAirportTerminal" : "odpt.AirportTerminal:NRT.Terminal2",
  "odpt:departureGate" : "85",
  "odpt:checkInCounter" : [ "B", "C" ],
  "odpt:destinationAirport" : "odpt.Airport:HKG",
  "odpt:aircraftType" : "788"
} ]
```

5.2.5. GET /api/v4/odpt:FlightSchedule

Description

Acquire the flight timetable (scheduled timetable for flights departing from and arriving at the airport).

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)
	odpt:operator	ID representing the airport operator or airline operator	string (odpt:Operator)

Query	<i>optional</i>	(owl:sameAs of odpt:Operator)	owl:sameAs)
Query	odpt:calendar <i>optional</i>	ID representing the day of operation (owl:sameAs of odpt:Calendar)	string (odpt:Calendar owl:sameAs)
Query	odpt:originAirport <i>optional</i>	ID representing the departure airport (owl:sameAs of odpt:Airport)	string (odpt:Airport owl:sameAs)
Query	odpt:destinationAirport <i>optional</i>	ID representing the destination airport (owl:sameAs of odpt:Airport)	string (odpt:Airport owl:sameAs)

Responses

HTTP Code	Description	Schema
200	Successful completion	< odpt:FlightSchedule > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Example HTTP request

Request path

```
/api/v4/odpt:FlightSchedule?acl:consumerKey=string
```

Example HTTP response

Response 200

```
[ {
  "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
  "@id" : "urn:ucode:_00001C0000000000000010000030FD7E5",
  "@type" : "odpt:FlightSchedule",
  "dc:date" : "2017-10-31T23:06:38+09:00",
  "owl:sameAs" : "odpt.FlightSchedule:HND-TIAT.HND.IWJ.Wednesday",
  "odpt:operator" : "odpt.Operator:HND-TIAT",
  "odpt:calendar" : "odpt.Calendar:Wednesday",
  "odpt:originAirport" : "odpt.Airport:HND",
  "odpt:destinationAirport" : "odpt.Airport:IWJ",
  "odpt:flightScheduleObject" : [ {
    "odpt:airline" : "odpt.Operator:ANA",
    "odpt:flightNumber" : [ "NH575" ],
    "odpt:originTime" : "10:35",
    "odpt:destinationTime" : "12:15",
    "odpt:isValidFrom" : "2017-10-01T00:00:00+09:00",
    "odpt:isValidTo" : "2017-10-31T23:59:59+09:00"
  } ]
} ]
```

Description

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>required</i>	Access token provided on the developer site.	string (acl:ConsumerKey)
Query	@id <i>optional</i>	Unique identifier assigned to data	string (URN)
Query	owl:sameAs <i>optional</i>	Alias of unique identifier assigned to data	string (URL)

HTTP Code	Description	Schema
200	Successful completion	< odpt:FlightStatus > array
400	Invalid parameter	No Content
401	acl:consumerKey is wrong.	No Content
403	Lacking authority	No Content
404	No applicable data	No Content
500	Server internal error	No Content
503	Service unavailable	No Content

Request path

```
/api/v4/odpt:FlightStatus?acl:consumerKey=string
```

Response 200

```
[ {
    "@context" : "http://vocab.odpt.org/context_odpt.jsonld",
    "@id" : "urn:ucode:_00001C0000000000000010000030FD7E5",
    "@type" : "odpt:FlightStatus",
    "dc:date" : "2017-01-13T15:10:00+09:00",
    "owl:sameAs" : "odpt.FlightStatus:CheckIn",
    "dc:title" : "搭乗手続中",
    "odpt:flightStatusTitle" : {
        "ja" : "搭乗手続中",
        "en" : "Check in"
    }
} ]
```

<https://developer.odpt.org/documents>

5.3.1. odpt:Airport

Represents airport information.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:Airport Example : <code>"odpt:Airport"</code>	string
dc:date <i>optional</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt:Airport:IATAairportcode (three letter code). Example : <code>"odpt.Airport:HND"</code>	string (URL)
dc:title <i>optional</i>	Airport name (Japanese). The name should be such that the location and the distinction from other airports can be clearly identified. Example : <code>"東京(羽田)"</code>	string
odpt:airportTitle <i>optional</i>	Airport name (multilingual-support) Example : <code>{"ja" : "東京(羽田)", "en" : "Tokyo (Haneda)"}</code>	object
odpt:airportTerminal <i>optional</i>	A list of IDs (owl:sameAs of odpt:AirportTerminal) representing airport terminals Example : <code>["odpt.AirportTerminal:HND.DomesticTerminal1"]</code>	< string (odpt:AirportTerminal owl:sameAs) > array
geo:long <i>optional</i>	Longitude of the representative location, decimal notation, geodetic system is WGS84 Example : <code>139.1234</code>	number
geo:lat <i>optional</i>	Latitude of representative location, decimal notation, geodetic system is WGS84 Example : <code>35.1234</code>	number
ug:region <i>optional</i>	Feature information in GeoJSON format Example : <code>"object"</code>	object

5.3.2. odpt:AirportTerminal

Represents airport terminal information.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C000000000000010000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:AirportTerminal Example : <code>"odpt:AirportTerminal"</code>	string
dc:date <i>optional</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)

owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.AirportTerminal.IATAairportcode (three letter code).ターミナル. When describing a wing, etc. in the same terminal, add .ウィング at the end. Example : "odpt.AirportTerminal:HND.DomesticTerminal1"	string (URL)
dc:title <i>optional</i>	Airport terminal name (Japanese) Example : "国内線第1ターミナル"	string
odpt:airportTerminalTitle <i>optional</i>	Airport terminal name (multilingual-support) Example : {"ja" : "国内線第1ターミナル", "en" : "Domestic Terminal 1"}	object
odpt:airport <i>required</i>	Airport ID (owl:sameAs of odpt:Airport) Example : "odpt.Airport:HND"	string (odpt:Airport owl:sameAs)
geo:long <i>optional</i>	Longitude of the representative location, decimal notation, geodetic system is WGS84 Example : 139.1234	number
geo:lat <i>optional</i>	Latitude of representative location, decimal notation, geodetic system is WGS84 Example : 35.1234	number
ug:region <i>optional</i>	Feature information in GeoJSON format Example : "object"	object

5.3.3. odpt:FlightInformationArrival

Flight arrival information represents real-time information on flights arriving at the airport on the given day.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : "http://vocab.odpt.org/context_odpt.jsonld"	string (URL)
@id <i>required</i>	Unique identifier (ucode or uuid) Example : "urn:uuid:ff91fee-2d2c-4365-902e-26375fdf9d8b"	string (URN)
@type <i>required</i>	ClassName name. odpt:FlightInformationArrival is returned here. Example : "odpt:FlightInformationArrival"	string
dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : "2017-12-07T13:10:00+09:00"	string (xsd:dateTime)
dct:valid <i>optional</i>	Data guarantee period (ISO8601 date and time format) Example : "2017-12-07T13:15:00+09:00"	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.FlightInformationArrival:Airportoperatororairlineoperator.Arrivalairport.Flightnumber. If duplicates occur with the above naming rules, they are distinguished by adding a suffix, .1, .2, .3, etc. Example : "odpt.FlightInformationArrival:HND-TIAT.HND.JL172"	string (URL)
odpt:operator <i>required</i>	ID representing the airport operator or airline operator providing the flight arrival information (owl:sameAs of odpt:Operator) Example : "odpt.Operator:HND-TIAT"	string (odpt:Operator owl:sameAs)
odpt:airline <i>optional</i>	ID representing the airline operator (owl:sameAs of odpt:Operator) Example : "odpt.Operator:ANA"	string (odpt:Operator owl:sameAs)
odpt:flightNumber <i>required</i>	A list of flight numbers. There will be more than one for codeshare flights. Example : ["NH105", "UA7983"]	< string > array

odpt:flightStatus <i>optional</i>	ID representing flight status (owl:sameAs of odpt:FlightStatus) Example : "odpt.FlightStatus:Arrived"	string (odpt:FlightStatus owl:sameAs)
odpt:flightInformationSummary <i>optional</i>	Summarizes information in natural language (multilingual-support) when there is information that needs special mention, such as when an operation failure occurs Example : {"ja" : "引き返し", "en" : "Turn back"}	object
odpt:flightInformationText <i>optional</i>	Describes information in natural language when there is information that needs special mention, such as when an operation failure occurs (multilingual-support) Example : {"ja" : "急病人救護のため引き返します", "en" : "Turn back for medical care"}	object
odpt:scheduledArrivalTime <i>optional</i>	Scheduled arrival time (ISO8601 time format) Example : "06:05"	string (odpt:Time)
odpt:estimatedArrivalTime <i>optional</i>	The arrival time after a change, (ISO8601 time format) and after arrival, odpt:actualArrivalTime is generated and odpt:estimatedArrivalTime cannot be acquired. Example : "06:04"	string (odpt:Time)
odpt:actualArrivalTime <i>optional</i>	The actual arrival time (ISO8601 time format), which does not exist until the flight arrives. Example : "06:06"	string (odpt:Time)
odpt:arrivalAirport <i>required</i>	ID representing the arrival airport (owl:sameAs of odpt:Airport) Example : "odpt.Airport:NRT"	string (odpt:Airport owl:sameAs)
odpt:arrivalAirportTerminal <i>optional</i>	ID representing the terminal at the arrival airport (owl:sameAs of odpt:AirportTerminal) Example : "odpt.AirportTerminal:NRT.Terminal2"	string (odpt:AirportTerminal owl:sameAs)
odpt:arrivalGate <i>optional</i>	Arrival airport gate number Example : "12A"	string
odpt:baggageClaim <i>optional</i>	Checked baggage claim area at the arrival airport Example : "12A"	string
odpt:originAirport <i>optional</i>	ID representing the departure airport (owl:sameAs of odpt:Airport) Example : "odpt.Airport:HKG"	string (odpt:Airport owl:sameAs)
odpt:viaAirport <i>optional</i>	A list of transfer airport IDs (owl:sameAs of odpt:Airport) Example : ["odpt.Airport:NRT"]	< string (odpt:Airport owl:sameAs) > array
odpt:aircraftType <i>optional</i>	Flight airplane model Example : "788"	string

5.3.4. odpt:FlightInformationDeparture

Flight departure information represents real-time information of the flights departing from the airport on the given day.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : "http://vocab.odpt.org/context_odpt.jsonld"	string (URL)
@id <i>required</i>	Unique identifier (ucode or uuid) Example : "urn:uuid:ff91feee-2d2c-4365-902e-26375fdf9d8b"	string (URN)
@type <i>required</i>	ClassName name. odpt:FlightInformationDeparture is returned here. Example : "odpt:FlightInformationDeparture"	string

dc:date <i>required</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-12-07T13:10:00+09:00"</code>	string (xsd:dateTime)
dct:valid <i>optional</i>	Data guarantee period (ISO8601 date and time format) Example : <code>"2017-12-07T13:15:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.FlightInformationDeparture:Airportoperatororairlineoperator.Departureairport.Flightnumber. If duplicates occur with the above naming rules, they are distinguished by adding a suffix, .1, .2, .3, etc. Example : <code>"odpt.FlightInformationDeparture:HND-TIAT.HND.JL172"</code>	string (URL)
odpt:operator <i>required</i>	ID of the airport operator or airline operator that provides the flight departure information (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:HND-TIAT"</code>	string (odpt:Operator owl:sameAs)
odpt:airline <i>optional</i>	ID representing the airline operator (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:ANA"</code>	string (odpt:Operator owl:sameAs)
odpt:flightNumber <i>required</i>	A list of flight numbers. There will be more than one for codeshare flights. Example : <code>["NH105", "UA7983"]</code>	< string > array
odpt:flightStatus <i>optional</i>	ID representing flight status (owl:sameAs of odpt:FlightStatus) Example : <code>"odpt.FlightStatus:Takeoff"</code>	string (odpt:FlightStatus owl:sameAs)
odpt:flightInformationSummary <i>optional</i>	Summarizes information in natural language, when there is any information that needs special mention (multilingual-support) Example : <code>{"ja" : "引き返し", "en" : "Turn back"}</code>	object
odpt:flightInformationText <i>optional</i>	Describes information in natural language when there is information that needs special mention (multilingual-support) Example : <code>{"ja" : "急病人救護のため引き返します", "en" : "Turn back for medical care"}</code>	object
odpt:scheduledDepartureTime <i>optional</i>	Scheduled departure time (ISO8601 time format) Example : <code>"06:05"</code>	string (odpt:Time)
odpt:estimatedDepartureTime <i>optional</i>	The departure time after a change, (ISO8601 time format), odpt:actualDepartureTime is generated after departure, and odpt:estimatedDepartureTime cannot be acquired. Example : <code>"06:04"</code>	string (odpt:Time)
odpt:actualDepartureTime <i>optional</i>	Actual departure time (time format of ISO8601), which does not exist until departure. Example : <code>"06:06"</code>	string (odpt:Time)
odpt:departureAirport <i>required</i>	ID representing the departure airport (owl:sameAs of odpt:Airport) Example : <code>"odpt.Airport:NRT"</code>	string (odpt:Airport owl:sameAs)
odpt:departureAirportTerminal <i>optional</i>	ID representing the terminal of the departure airport (owl:sameAs of odpt:AirportTerminal) Example : <code>"odpt.AirportTerminal:NRT.Terminal2"</code>	string (odpt:AirportTerminal owl:sameAs)
odpt:departureGate <i>optional</i>	Departure airport gate number Example : <code>"12A"</code>	string (xst:string)
odpt:checkInCounter <i>optional</i>	A list of departure airport check-in counters Example : <code>["A", "B"]</code>	< string > array
odpt:destinationAirport <i>optional</i>	ID representing the destination airport (owl:sameAs of odpt:Airport) Example : <code>"odpt.Airport:HKG"</code>	string (odpt:Airport owl:sameAs)

odpt:viaAirport <i>optional</i>	A list of transfer airport IDs (owl:sameAs of odpt:Airport) Example : <code>["odpt:Airport:NRT"]</code>	< string (odpt:Airport owl:sameAs) > array
odpt:aircraftType <i>optional</i>	Flight airplane model Example : <code>"788"</code>	string

5.3.5. odpt:FlightSchedule

Flight timetables represent scheduled timetables for flights departing from and arriving at the airport.

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000001000030FD7E5"</code>	string (URN)
@type <i>required</i>	ClassName name, odpt:FlightSchedule is returned. Example : <code>"odpt:FlightSchedule"</code>	string
dc:date <i>optional</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-12-07T13:10:00+09:00"</code>	string (xsd:date)
owl:sameAs <i>required</i>	Unique identifier. The naming rules are: odpt.FlightSchedule:Airportoperatororairlineoperator.Airportoforigin.Airportofdestination.Calendarinformation. Example : <code>"odpt.FlightSchedule:ANA.HND.ITM.Wednesday"</code>	string (URL)
odpt:operator <i>required</i>	ID representing the airport operator or airline operator that provides the flight timetable (owl:sameAs of odpt:Operator) Example : <code>"odpt.Operator:ANA"</code>	string (odpt:Operator owl:sameAs)
odpt:calendar <i>required</i>	ID representing calendar information of the day when the flight timetable is applicable (owl:sameAs of odpt:Calendar) Example : <code>"odpt.Calendar:Wednesday"</code>	string (odpt:Calendar owl:sameAs)
odpt:originAirport <i>required</i>	ID of the departure airport (owl:sameAs of odpt:Airport) Example : <code>"odpt.Airport:HND"</code>	string (odpt:Airport owl:sameAs)
odpt:destinationAirport <i>required</i>	ID of the destination airport (owl:sameAs of odpt:Airport) Example : <code>"odpt.Airport:ITM"</code>	string (odpt:Airport owl:sameAs)
odpt:flightScheduleObject <i>optional</i>	Airflight timetable object Example : <code>["object"]</code>	< odpt:FlightScheduleObject > array

odpt:FlightScheduleObject

Name	Description	Schema
odpt:airline <i>required</i>	Airline operator ID Example : <code>"odpt.Operator:ANA"</code>	string (odpt:Operator owl:sameAs)
odpt:flightNumber <i>required</i>	A list of flight numbers. There will be more than one for codeshare flights. Example : <code>["NH105", "UA7983"]</code>	< string > array
odpt:originTime <i>required</i>	Scheduled departure time from the departure airport (ISO8601 time format, local time) Example : <code>"10:00"</code>	string (odpt:Time)

odpt:originDayDifference <i>optional</i>	The difference between departure date and calendar information (odpt:calendar of odpt:FlightSchedule). Example: 0 means departure on the same day, 1 means departure on the next day, -1 means departure on the previous day Example : <code>0</code>	integer
odpt:destinationTime <i>required</i>	Estimated time of arrival at the destination airport (ISO8601 time format, local time) Example : <code>"12:15"</code>	string (odpt:Time)
odpt:destinationDayDifference <i>optional</i>	The difference between the arrival date and the calendar information (odpt:calendar of odpt:FlightSchedule). Example: 0 means arrival on the same day, 1 means arrival on the next day, -1 means arrival on the previous day Example : <code>1</code>	integer
odpt:viaAirport <i>optional</i>	A list of transfer airport IDs (owl:sameAs of odpt:Airport) Example : <code>["odpt:Airport:NRT"]</code>	< string (odpt:Airport owl:sameAs) > array
odpt:aircraftType <i>optional</i>	Flight airplane model Example : <code>"788"</code>	string
odpt:isValidFrom <i>optional</i>	Data is application from the start date and time (ISO8601 date and time format) Example : <code>"2017-10-31T23:06:38+09:00"</code>	string (xsd:dateTime)
odpt:isValidTo <i>optional</i>	Data is applicable until the end date and time (ISO8601 date and time format) Example : <code>"2017-10-31T23:06:38+09:00"</code>	string (xsd:dateTime)
odpt:note <i>optional</i>	Natural language description of annotation information not defined as properties (multilingual-support) Example : <code>{"ja" : "日本語での注釈情報", "en" : "Note in English"}</code>	object

5.3.6. odpt:FlightStatus

Defines the status of flights arriving and departing at the airport, such as boarding and departed

Name	Description	Schema
@context <i>required</i>	@context URL format in JSON-LD Example : <code>"http://vocab.odpt.org/context_odpt.jsonld"</code>	string (URL)
@id <i>required</i>	Unique identifier (ucode) Example : <code>"urn:ucode:_00001C0000000000001000030FD7E5"</code>	string (URN)
@type <i>required</i>	Name of the className. odpt:FlightStatus Example : <code>"odpt:FlightStatus"</code>	string
dc:date <i>optional</i>	Data generation date and time (ISO8601 date and time format) Example : <code>"2017-01-13T15:10:00+09:00"</code>	string (xsd:dateTime)
owl:sameAs <i>required</i>	Unique identifier. The naming rule is odpt.FlightStatus:flightStatus. Example : <code>"odpt.FlightStatus:CheckIn"</code>	string (URL)
dc:title <i>optional</i>	Flight status (Japanese) Example : <code>"搭乗手続中"</code>	string
odpt:flightStatusTitle <i>optional</i>	Flight status (multilingual-support) Example : <code>{"ja" : "搭乗手続中", "en" : "Check in"}</code>	object

6. ODPT File API

6.1. Overview

API for acquiring published files

6.1.1. Version information

Version : 1.0

6.1.2. URI scheme

Host : api.odpt.org

BasePath : /v4/files

Schemes : HTTPS

6.2. Paths

6.2.1. GET /v4/files/

Description

API for acquiring files published by each company

Parameters

Type	Name	Description	Schema
Query	acl:consumerKey <i>optional</i>	subscription key in url	string
Query	path <i>required</i>	path to resource	enum (Dictionary.yaml)

Example HTTP request

Request path

```
/v4/files/?path=string
```

7. Appendix :: About RDF and JSON-LD

This section describes the technologies used in public transportation open data, and introduces tips for applications.

7.1. URI (Uniform Resource Identifier)

A URI (Uniform Resource Identifier) is a character string for uniquely specifying a resource on the Internet, and can be classified into a URL (Uniform Resource Locator) and a URN (Uniform Resource Name).

A URL uniquely identifies a resource on the Internet by combining scheme name (HTTP, FTP, etc.), host name, path name, etc. in the proper order. Because you specify a resource by specifying a host name or path, it can be compared to the Internet address of the resource. You are probably familiar with URIs that are used when browsing websites.

A URN uniquely identifies a resource on the Internet by combining scheme name, namespace identifiers, and namespace-specific string. Since an identifier is used to specify a resource, it can be compared to the name of the resource on the Internet. As examples of URN, ucode (Ubiquitous Code) issued by uID Center and ISBN (International Standard Book Number) assigned to books are used.

Conventionally, characters that can be used in URIs are limited to ASCII characters, but today, IRIs (Internationalized Resource Identifiers), which are extended URIs to handle many languages including Japanese, are widely used. The term URI used in the explanation in this document may be replaced with IRI as it is, but we use the term URI from the viewpoint of understandability and familiarity.

7.2. RDF (Resource Description Framework)

RDF (Resource Description Framework) is a standard created by the W3C for exchanging data on the Web. Traditionally, information posted on the Web was written so that humans could read and understand the content, but RDF was designed to enable computers to interpret the content of resources on the Internet (machine-readable). RDF is designed to handle not only resources on the Internet, but also objects in the real world, concepts, and character strings.

One RDF sentence consists of a subject representing the subject of explanation, a predicate representing an attribute of the subject, and an object representing an attribute value. A series of multiple RDF sentences is called an RDF graph, and is expressed as a directed graph in which nodes representing subjects and nodes representing objects are connected by arrows representing predicates (Fig. 1). A subject stores a URI or a blank node (a node representing null), the predicate stores a URI, and the object stores a URI, a literal (a character string or a number), or a blank node. The URI stored in the predicate is generally called a property because it represents the attribute of the subject, and the URI and literal are collectively called a resource. Examples of actual applications of RDF include RSS (RDF Site Summary) and OGP (Open Graph Protocol), which is used to display an overview of link destinations on SNS (Social Networking Service) such as Facebook. Usage is increasing.

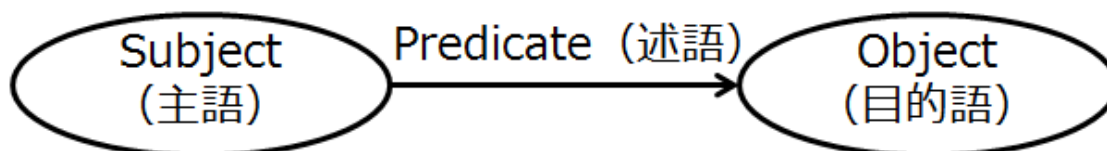
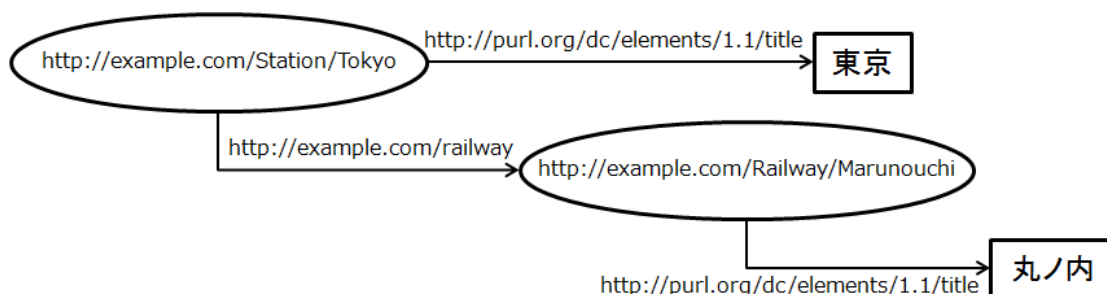


Figure 2 shows an RDF statement in which the title (<http://purl.org/dc/elements/1.1/title>) of the resource represented by <http://example.com/Stations/Tokyo> is "東京". The route (<http://example.com/railway>) of the resource represented by: <http://example.com/Stations/Tokyo> is <http://example.com/Railway/Marunouchi>, and its title is "丸ノ内". This is an RDF graph that combines RDF sentences representing that circles in the figure represent URIs, boxes represent literals, and arrows represent properties.



7.3. RDF Schema

RDF describes the relationship among resources. Similarly, properties that represent the relationship describe the usage and meaning of properties using a vocabulary called RDF Schema. RDF Schema has the concept of a `className`, similar to an object-oriented programming language. A `className` represents a type of resource, and a resource belonging to a certain `className` is called an instance of a certain `className`. For example, instances of the railway station `className` include Tokyo Station and Shinjuku Station. RDF Schema defines a vocabulary for expressing the `className` that can be the subject of a property and the `className` that can be the object of a property in the RDF graph together with the meaning of the property itself.

7.4. Vocabulary

A vocabulary is a collection of `className`s and properties that can be used to describe RDF graphs, and is described using RDF Schema or the like. There are vocabularies such as RDF Schema designed to define RDF schemas themselves, Dublin core designed to describe resource metadata, and FOAF (Friend of a Friend) designed

to describe information about people.

The vocabulary definition includes RDF statements that starts with the same URI. For example, the vocabulary of the RDF Schema includes an RDF graph about the subject starting with `http://www.w3.org/2000/01/rdf-schema#`, such as `http://www.w3.org/2000/01/rdf-schema#ClassName` representing the `className` and `http://www.w3.org/2000/01/rdf-schema#comment` representing the description. The common part of this URI is called a namespace. When a vocabulary is used, a prefix which is an abbreviation of this namespace is used. The prefix and the URI after the prefix are described by connecting them with a colon. For example, RDF Schema `http://www.w3.org/2000/01/rdf-schema#` uses the abbreviation `rdfs`. The resources representing the `className` and the description described above can be expressed in a very simple format, `rdfs:ClassName` and `rdfs:comment`, respectively.

7.5. Description of RDF

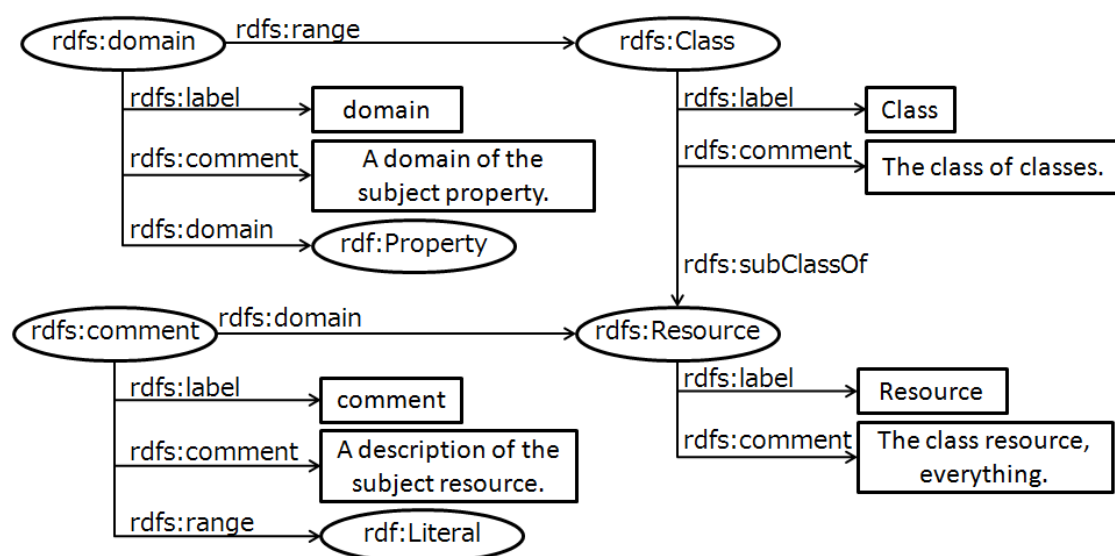
Defining vocabulary and describing data are generally done via RDF / XML format using XML format, Turtle format which is a lightweight format for RDF description, and JSON-LD format using JSON format. For example, `rdfs:ClassName` which is a `className` representing a `className`, `rdfs:Resource` that is a `className` representing a resource, `rdfs:domain` representing a `className` that can be the subject of a property, and the definition of the vocabulary `rdfs:comment` are excerpted from the RDFS (RDF Schema) definition file. The generated tables (Table 1 and Table 2) and the RDF graph (Figure 3) are shown, respectively.

Table 1: ClassName definitions

URI	ClassName name (rdfs:label)	Description (rdfs:comment)
<code>rdfs:ClassName</code>	<code>ClassName</code>	The <code>className</code> of <code>className</code> es.
<code>rdfs:Resource</code>	<code>Resource</code>	The <code>className</code> resource, everything.

Table 2: Property definitions

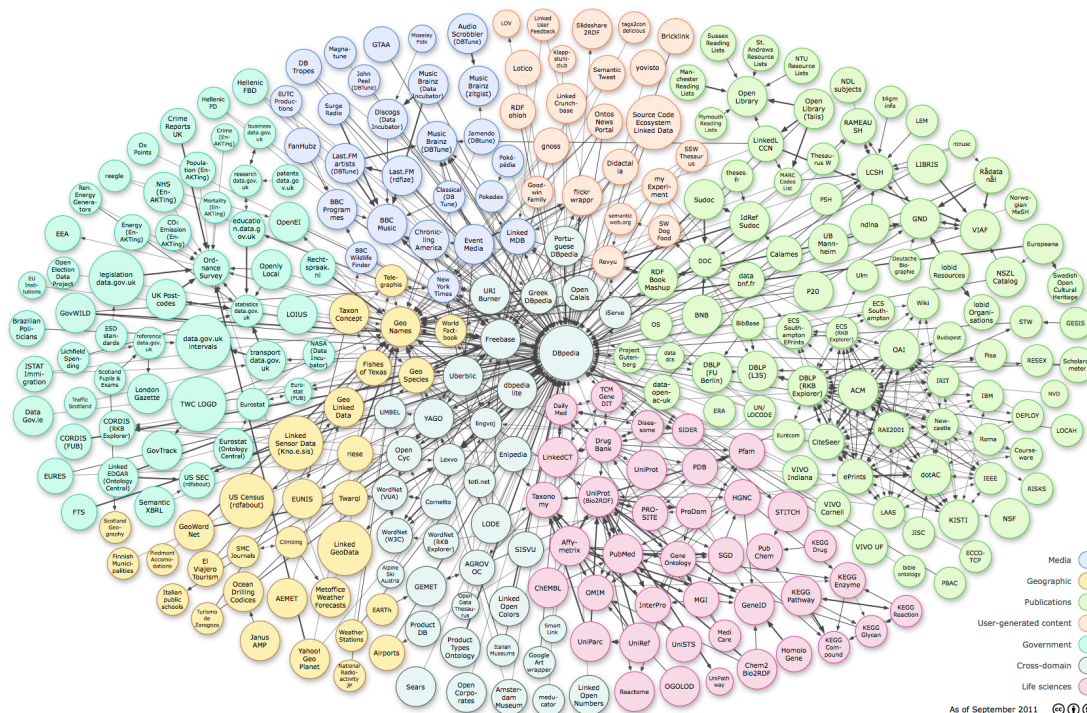
URI	Property name (rdfs:label)	Description (rdfs:comment)	Subject className (rdfs:domain)	Predicate className (rdfs:range)
<code>rdfs:domain</code>	<code>domain</code>	A domain of the subject property.	<code>rdf:Property</code>	<code>rdfs:ClassName</code>
<code>rdfs:comment</code>	<code>comment</code>	A description of the subject resource.	<code>rdfs:Resource</code>	<code>rdfs:Literal</code>



7.6. Linked Data

Linked Data describes the relationship between data described using RDF and other data already described and

published in RDF, etc. in a certain way. This description links various resources in the world to make it easier for people and computers to discover new data. As the relation to be described, owl:sameAs representing that the subject and the object are identical, and skos:related representing that the subject and the object are related, and others are used. Various organizations and individuals publish Linked Data, but the largest released data is DB Pedia, the Wikipedia turned into Linked Data. Various LinkedData have already been published around the world, and Fig. 4 is famous, where the number of data is represented by the size of a circle and the number of links is represented by a line thickness. (Linking Open Data cloud diagram, by Richard Cyganiak and Anja Jentzsch.)



7.7. JSON-LD

JSON-LD (JSON for Linked Data) is a format formulated to describe Linked Data using the JSON format widely adopted in web application development, and became a W3C recommendation on January 16, 2014. Therefore, it is expected to be widely used in the future. The main features are as follows.

- Compatible with normal JSON
 - Since the value can be fetched using a normal JSON parser, the same data can be handled as just JSON data or as Linked Data depending on the purpose. For this reason, the data is easy to handle even when it is not necessary to handle it as Linked Data. Many applications created using this API do not need to interpret the data as RDF, so it would be convenient to be able to handle it as normal JSON data.
- Properties for a subject can be described collectively as one JSON object.
 - A subject is defined by the key named @id in the JSON object, and properties for the subject can be described collectively in the same object, so that readability can be improved, and the amount of description can be reduced.
- Prefix can be defined.
 - A key named @@context is defined, and by describing the prefix and URI of the namespace in this key, the abbreviation that uses the prefix can be used to improve readability and reduce the amount of description.
- Nesting of RDF statements can be described.
 - The RDF statement may be a nested graph as shown in Fig. 2. It can be described in JSON-LD in the same way that objects can be nested in normal JSON.

As a description sample, an example is given where the vocabulary definitions described in Tables 1 and 2 are expressed in JSON-LD format. Note that this document does not explain the JSON-LD specification itself, so please

refer to JSON-LD specifications by W3C for detailed specification. Also, the library for handling JSON-LD as RDF in each programming language is introduced on the official website of JSON-LD, so please refer to it as well.

```
{
  "@context": {
    "rdfs": "http://www.w3.org/2000/01/rdf-schema#",
  },
  "@id": "http://www.w3.org/2000/01/rdf-schema#",
  "@graph": [
    {
      "@id": "rdfs:Resource",
      "@type": "rdfs:ClassName",
      "rdfs:label": "Resource",
      "rdfs:comment": "The className resource, everything."
    },
    {
      "@id": "rdfs:ClassName",
      "@type": "rdfs:ClassName",
      "rdfs:label": "ClassName",
      "rdfs:comment": "The className of classNameees."
    },
    {
      "@id": "rdfs:comment",
      "@type": "rdfs:Property",
      "rdfs:label": "comment",
      "rdfs:comment": "A description of the subject resource.",
      "rdfs:domain": "rdfs:Resource",
      "rdfs:range": "rdfs:Literal"
    },
    {
      "@id": "rdfs:domain",
      "@type": "rdfs:Property",
      "rdfs:label": "domain",
      "rdfs:comment": "A domain of the subject property.",
      "rdfs:domain": "rdfs:ClassName",
      "rdfs:range": "rdf:Property"
    }
  ]
}
```

8. Update history

2024-03-25

- Added `odpt:connectingStation` (transferable stations) to `odpt:Station` (station information).

2023-11-09

- Added `odpt:platformNumber` (bus stop platform number) to `odpt:BusstopPole` (bus stop information).

2022-09-20

- Changed data type of `odpt:operator` (operator ID) of `odpt:BusroutePattern` (bus route) from a list (array) to a string.

2021-08-23

- Added a description, when more than one specific `classNameees` are applicable to the same date, to `odpt:Calendar` (day of the week / date category).
- Added `odpt:busroutePatternOrder` (order of bus stops (posts) visited in the route pattern) to `odpt:BusstopPoleTimetable` (bus stop (post) timetable).

- Other typographical errors have been corrected.

2021-03-04

- Added `odpt:occupancyStatus` (occupancy status) to `odpt:Bus` (bus information).

2021-02-03

- Added `odpt:busTimetable` (timetable ID) to `odpt:Bus` (bus information).

2020-06-02

- Added a description about the data types, supported by the data acquisition API (GET `/v4/datapoints/$DATA_URI`).

2019-08-08

- Modified the description of the train numbers used in `odpt:Train` (train information) etc. because there are cases where ODPT assigns their own train numbers.
- `odpt:Train` (train information) represents the train location (position), but it does not distinguish whether a train is stopped or passing through a station, so the description was modified.
- In `odpt:Bus` (Bus Operation Information), added a description about the case where it is not possible to distinguish whether a bus is stopped or approaching a bus stop.
- In `odpt:BusstopPole` (bus stop information), title (multilingual bus stop name) was added. (For `dc:title`, only Japanese name was used as usual, and multilingual support was provided with the 'title' of another property.)
- In addition, the format of the specification has been improved somewhat.

2019-06-07

- Fixed typographical errors.
 - The response example of the data dump API (`/v4/RDF_TYPE.json?`) has been corrected.
 - `ug:region` (feature information) of `odpt:BusroutePattern` (bus route) and `odpt:BusstopPole` (bus stop) has been modified to GeoJSON object.

2019-04-25

- We can now specify object type key for data search filtering API. In addition, the handling of array type is now described.
- Clarified that the reference for latitude and longitude is WGS84 (World Geodetic System).
- The description of the priority of calendars when there are more than one calendar, `odpt:Calendar` (day of the week, date), for a given date has been corrected.
- Added `dct:issued` (revision date) to `odpt:TrainTimetable` (train timetable), `odpt:StationTimetable` (station timetable), `odpt:RailwayFare` (railway fare), `odpt:BusTimetable` (bus timetable), `odpt:BusstopPoleTimetable` (bus stop (post) timetable), and `odpt:BusroutePatternFare` (bus fare).
- Added `dct:valid` (guarantee period) to `odpt:TrainTimetable` (train timetable), `odpt:StationTimetable` (station timetable), `odpt:RailwayFare` (railway fare), `odpt:FlightInformationArrival` (flight arrival information), and `odpt:FlightInformationDeparture` (flight departure information).
- Added `odpt:ascendingRailDirection` and `odpt:descendingRailDirection` (ascending / descending direction) to `odpt:Railway` (railway line).
- Added description of how to handle loop lines etc. to `odpt:StationOrder` (station order) of `odpt:Railway` (railway line).
- Added `odpt:includeAlighting` (distinguishing between passengers getting on and off and passengers getting on) to `odpt:PassengerSurvey` (number of passengers getting on and off at the train station).
- Changed applied direction of `odpt:TrainInformation` (train operation information) from `odpt:trainInformationLine` to `odpt:railDirection`.
- Added description about the location of `odpt:Bus` (bus operation information).
- Added `odpt:originDayDifference` (day difference of departure date), `odpt:destinationDayDifference` (day difference of arrival date), and `odpt:note` (annotation) to `odpt:FlightScheduleObject` (object of flight timetable).

2018-09-13

- Added `odpt:viaRailway` (via / transfer line) and `odpt:viaStation` (via / transfer station) to `odpt:RailwayFare` (railway fare). If the fare is different for each route, it is used to represent which route the fare is for.
- Added `odpt:Airport` (Airport Information), `odpt:AirportTerminal` (Airport Terminal Information), and `odpt:FlightStatus` (Flight Status).
- In `odpt:FlightInformationArrival` (flight arrival information), `odpt:FlightInformationDeparture` (flight departure information), and `odpt:FlightSchedule` (flight timetable), the changes of properties and other were made. Programs that conform to the old specification require modification. An overview of the changes is as follows.
 - `odpt:FlightInformationArrival`
 - Changed property name: `odpt:scheduledTime` -> `odpt:scheduledArrivalTime`, `odpt:estimatedTime` -> `odpt:estimatedArrivalTime`, `odpt:actualTime` -> `odpt:actualArrivalTime`, `odpt:destinationAirport` -> `odpt:arrivalAirport`, `odpt:departureAirport` -> `odpt:originAirport`, `odpt:terminal` -> `odpt:arrivalAirportTerminal`
 - Changed value: `owl:sameAs`, `odpt:airline` (`odpt:Airline` -> `odpt:Operator`), `odpt:baggageClaim`
 - Changed property name and value: `odpt:gate` -> `odpt:arrivalGate`
 - Added properties: `odpt:flightInformationSummary`, `odpt:flightInformationText`, `odpt:viaAirport`, `odpt:aircraftType`
 - `odpt:FlightInformationDeparture`
 - Rename property: `odpt:scheduledTime` -> `odpt:scheduledDepartureTime`, `odpt:estimatedTime` -> `odpt:estimatedDepartureTime`, `odpt:actualTime` -> `odpt:actualDepartureTime`, `odpt:terminal` -> `odpt:departureAirportTerminal`
 - Changed the value: `owl:sameAs`, `odpt:airline` (`odpt:Airline` -> `odpt:Operator`)
 - Changed property name and value: `odpt:gate` -> `odpt:departureGate`
 - Added properties: `odpt:flightInformationSummary`, `odpt:flightInformationText`, `odpt:checkInCounter`, `odpt:viaAirport`, `odpt:aircraftType`
 - `odpt:FlightSchedule`, `FlightScheduleObject`
 - Renamed property: `odpt:departureTime` -> `odpt:originTime`, `odpt:arrivalTime` -> `odpt:destinationTime`
 - Changed values: `owl:sameAs`, `odpt:airline` (`odpt:Airline` -> `odpt:Operator`), `odpt:flightNumber` (now an array)
 - Renamed property and moved property: `odpt:departureAirport` (in `FlightScheduleObject`) -> `odpt:originAirport` (in `FlightSchedule`)
 - Moved property: `odpt:destinationAirport` (in `FlightScheduleObject`) -> `odpt:destinationAirport` (in `FlightSchedule`)
 - Added property: `odpt:viaAirport`, `odpt:aircraftType`
- Other typographical errors have been corrected.

2018-08-09

- `odpt:Calendar` (day of the week / date), `odpt:Operator` (operator), `odpt:TrainType` (train type), `odpt:RailDirection` (train direction) are also described in data dump API.
- The description of the feature information search API filtering (`/v4/places/RDF_TYPE?`) was modified.
- In `odpt:Train` (train information), `odpt:TrainTimetable` (train timetable) and `odpt:StationTimetableObject` (station timetable object), the `className` of ID of `odpt:trainOwner` (train owner) has become `odpt:Operator` (operator). It is used when there is a possibility that the operator and the company to which the vehicle belongs may be different due to mutual entry.
- In `odpt:TrainTimetable` (train timetable) and `odpt:StationTimetableObject` (station timetable object), changed `odpt:trainName` (train name) to a list (array). This is to support a joined train with multiple train names.
- Added `odpt:trainNumber` (train number) to `odpt:StationTimetableObject` (station timetable object).
- In `odpt:FlightInformationArrival` (flight arrival information) and `odpt:FlightInformationDeparture` (flight departure information), changed `odpt:flightNumber` (flight number) to a list (array). This is for code-sharing flights with multiple flight numbers.

- Other typographical errors have been corrected.

2018-07-17

The main modifications compared to the specification as of January 2018 are as follows.

- Added description of API to acquire information of `odpt:Calendar` (day of the week / date), `odpt:Operator` (operator), `odpt:TrainType` (train type), `odpt:RailDirection` (train traveling direction).
- Various character strings in natural languages related to railway are now multilingual. They are expressed in the JSON-LD language map (e.g. `{"ja": "日本語", "en": "English"}`). However, `dc:title` is limited to Japanese only as before, and multilingual support is provided by another property.
- In `odpt:TrainTimetable` (train timetable) and `odpt:StationTimetable` (station timetable), changed `odpt:originStation` (starting station) and `odpt:destinationStation` (terminal station) from string to list (array). This is to cope with joined trains that have multiple destinations because they are split on the way.
- In `odpt:TrainTimetable` (train timetable), added `odpt:previousTrainTimetable` (preceding train timetable) and `odpt:nextTrainTimetable` (succeeding train timetable). This is to deal with the case where one train is listed in multiple train timetables. In the case of a joined train that is split on the way, the train timetable immediately before or after may be plural, so it is a list (array).
- Changed the specification of `odpt:PassengerSurvey` (the number of passengers getting on and off at the station). This is to deal with the case where the statistics of the number of passengers getting on and off are divided for different lines at the same station name.

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